origins-of-life-vent · investigation report

# Origins-of-life vent campaign

running · viva-genesis · 10 studies · 2026-10-11

## Executive summary

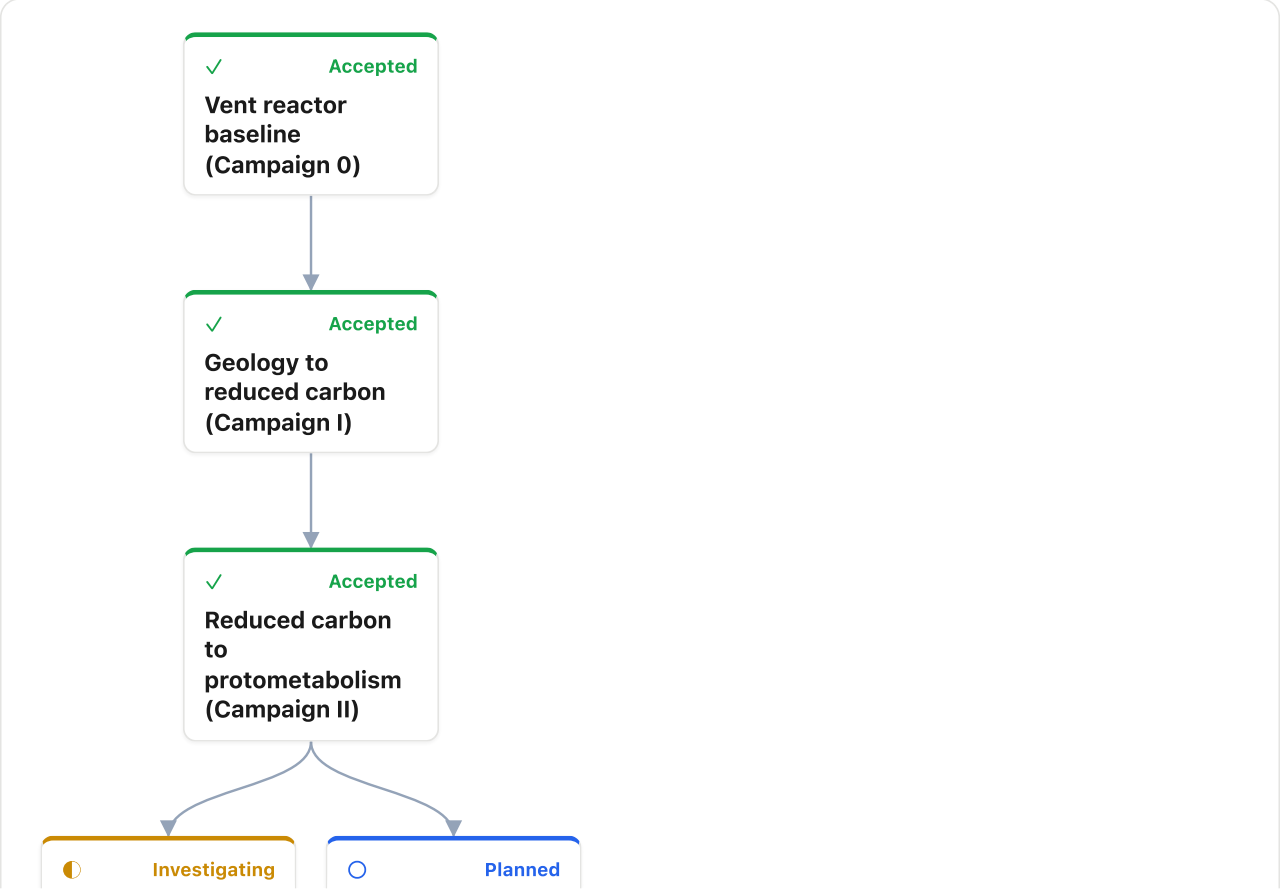
**Question.** Could the first cells have originated at alkaline hydrothermal vents? Can a physically plausible early-Earth vent–ocean interface generate and sustain, in order, the free energy, reduced carbon, self-amplifying chemistry, concentration, and compartments a protocell needs — without inserting metabolism, membranes, or heredity by assumption?

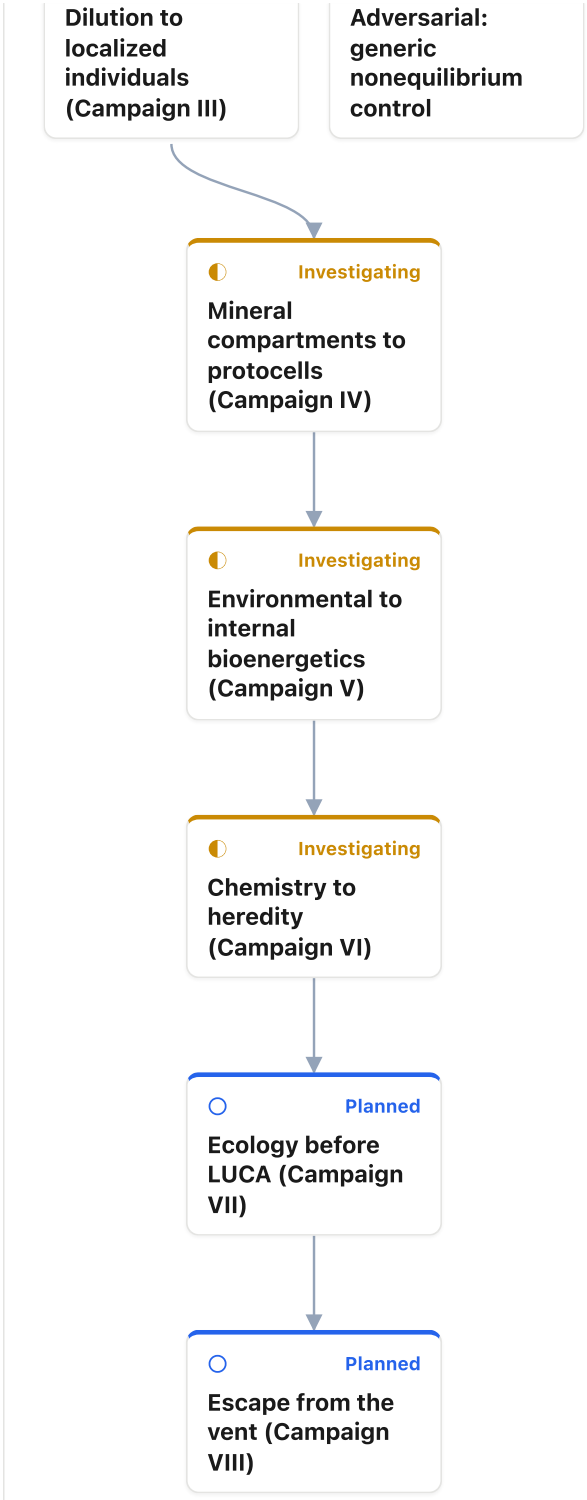
In progress — the pre-genetic milestone is three-for-three. og-00 (geochemical/spatial testbed), og-01 (geology → reduced carbon, favorable vs. Tutolo-2026 arms) and og-02 (reduced carbon → protometabolism) are built on real simulators, run, and accepted against pre-stated gates with conservation held to machine precision. og-03 (dilution → localized individuals) is next; og-04–og-08 and the adversarial control are designed but not yet run.

Across **10** studies — **6** supported · **0** qualified · **0** refuted · **3** planned. *Supported* = a computational gate passed under the stated model assumptions (evidence of feasibility, not proof the transition occurred).

## The investigation graph

The campaign as a dependency graph — each study a node badged with its code-computed status, wired to the studies it builds on.






Status: ✓ Accepted ◐ Investigating ○ Planned ⊘ Blocked ✕ Refuted

## Studies

Expand all | Collapse all

Every study in the campaign as a collapsible card — the question it tests, its verdict, and what it found at a glance. Click a card to expand the full record: model, runs, figures, tests, findings, and the limitations that bound it.

### ▾ 01 Vent reactor baseline (Campaign 0)

ACCEPTED

*Can a Lost City–like, serpentinization-fed vent–ocean interface reproduce observed fluid mixing, heat transport, mineral precipitation and H2/alkalinity delivery under field-constrained conditions — with no origin-of-life*

*claim?*

**Hypothesis & discriminating control**

A 1D/2D porous-wall reactor with prescribed boundary fluxes reproduces published Lost City ranges (T, pH, H2, DIC) and qualitative precipitation trends while conserving mass and charge.

MODEL EXPERIMENT Build a reduced vent–ocean reactor and validate it against Lost City field ranges to establish a geochemical/spatial testbed.

OBSERVABLE `temperature, pH, H2_flux, charge_imbalance, flux_divergence_residual`

**Model & key assumptions**

**Methods · Simulations**

| | |
|---|---|
| Model | `serpentinizing_vent_reactor` |
| Sims | 6 runs · completed |
| Tests | ✓✓✓✓✓ 5/5 passed |

KEY ASSUMPTIONS

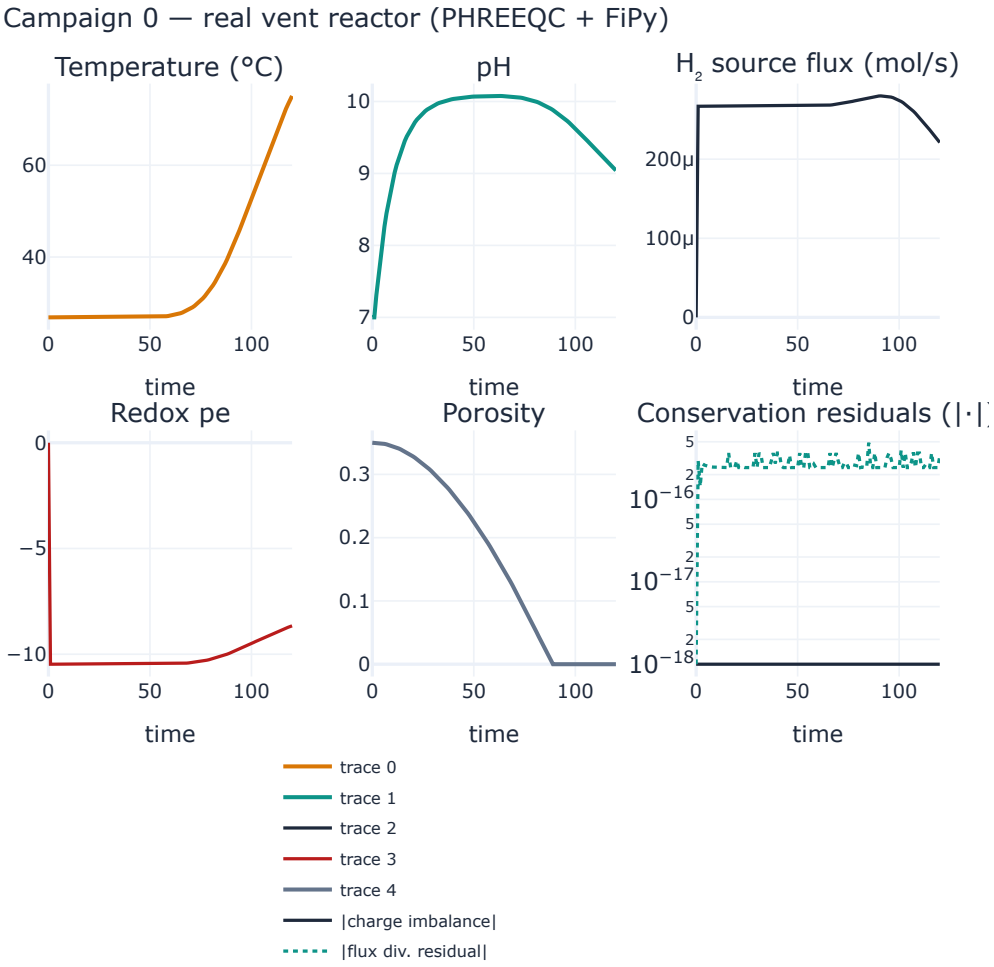
- Hadean ocean chemistry parameterized separately — never reuse modern seawater as the ancient prior.
- Partial equilibrium: a prescribed olivine dissolution rate (flow/T-driven) feeds PHREEQC equilibrium speciation each step — fast aqueous/mineral equilibrium vs a rate-limited rock reaction, standard for reactive-transport.
- The heat-partition fraction (fraction of serpentinization enthalpy retained in the fluid parcel) is CALIBRATED to the observed 40-91 C vent temperature; the chemistry (H2, pH, pe, charge) is emergent from PHREEQC, not tuned.

**Visualizations**

BEHAVIOR TESTS

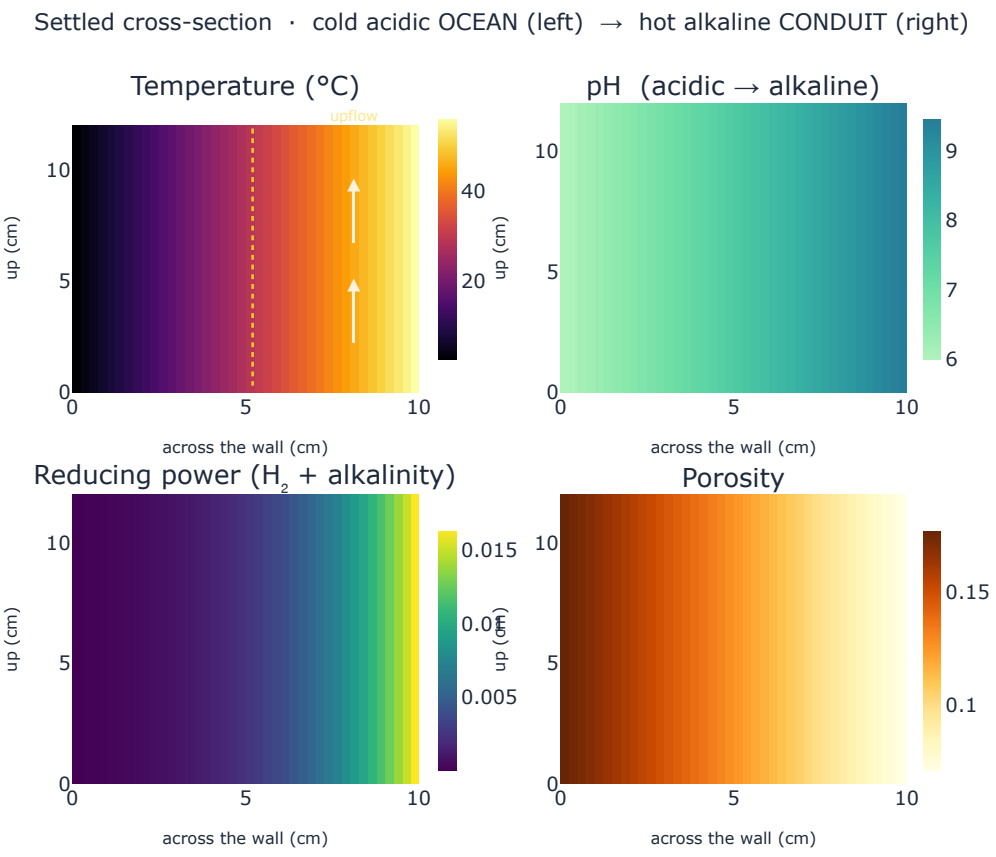
CONCLUSION

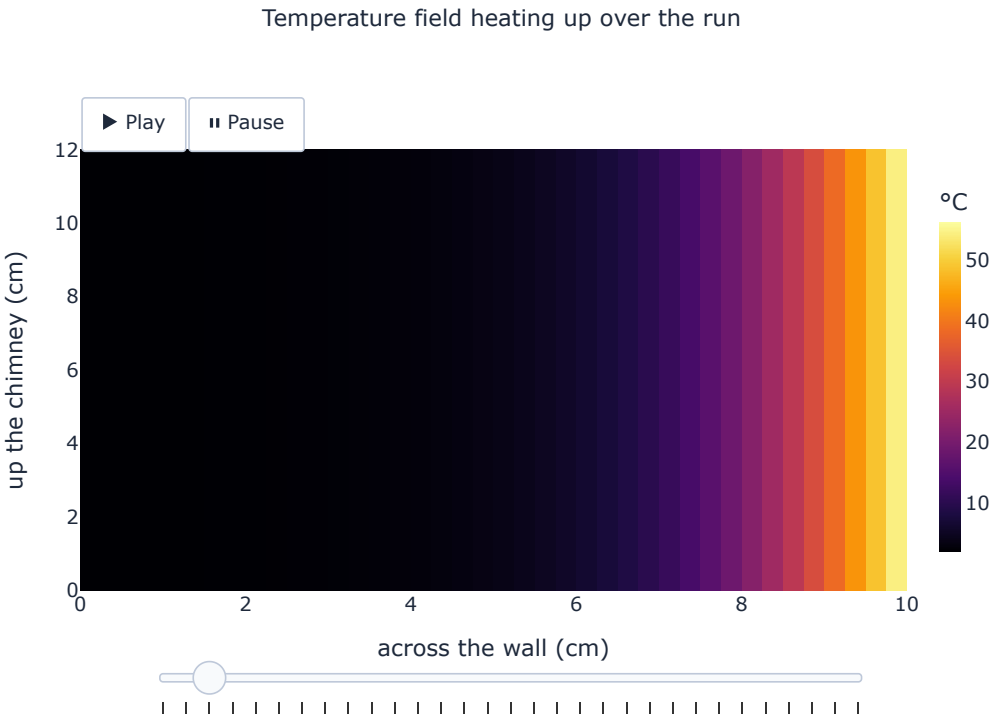
VENT REACTOR TIMESERIES



VENT CROSS SECTION

**A cross-section through the chimney wall** — a real 2-D FiPy finite-volume field, **every pixel a solved cell**. Cold acidic seawater bathes the outer wall (left); the hot alkaline serpentinized fluid percolates up the conduit (right). The **gradient across the wall** is the engine of the vent. **Settled: 49 °C, pH 9.45** (the Lost City window).




**Finding**

og00-temperature-ph-in-window · confirms · Observed

Outlet settles at 59.2 °C and pH 9.46 — both inside the Lost City field window (40–91 °C, 9.0–10.8).
Kelley 2001 Ludwig 2006

from test: vent-temperature-in-lostcity-range `range` BAND Kelley 2001 Ludwig 2006

og00-h2-sustained · confirms · Observed

Serpentinization delivers a sustained H2 reducing-power flux (min 2.2e-4 mol/s over the back half of the run).
Schwander 2023

from test: h2-flux-sustained > 1e−9 BAND Schwander 2023

og00-conservation · confirms · Observed

PHREEQC charge balance and FiPy FVM spatial-flux divergence are conserved to machine epsilon (<1e-6 and ~5e-16).

from test: charge-conserved <= `0.000001`

og00-peridotite-endmember · novel · Plausible

The Lost City pH endmember required reacting PERIDOTITE (olivine+pyroxene), not dunite — Ca released once Mg is consumed buffers the fluid to pH ~9.5.

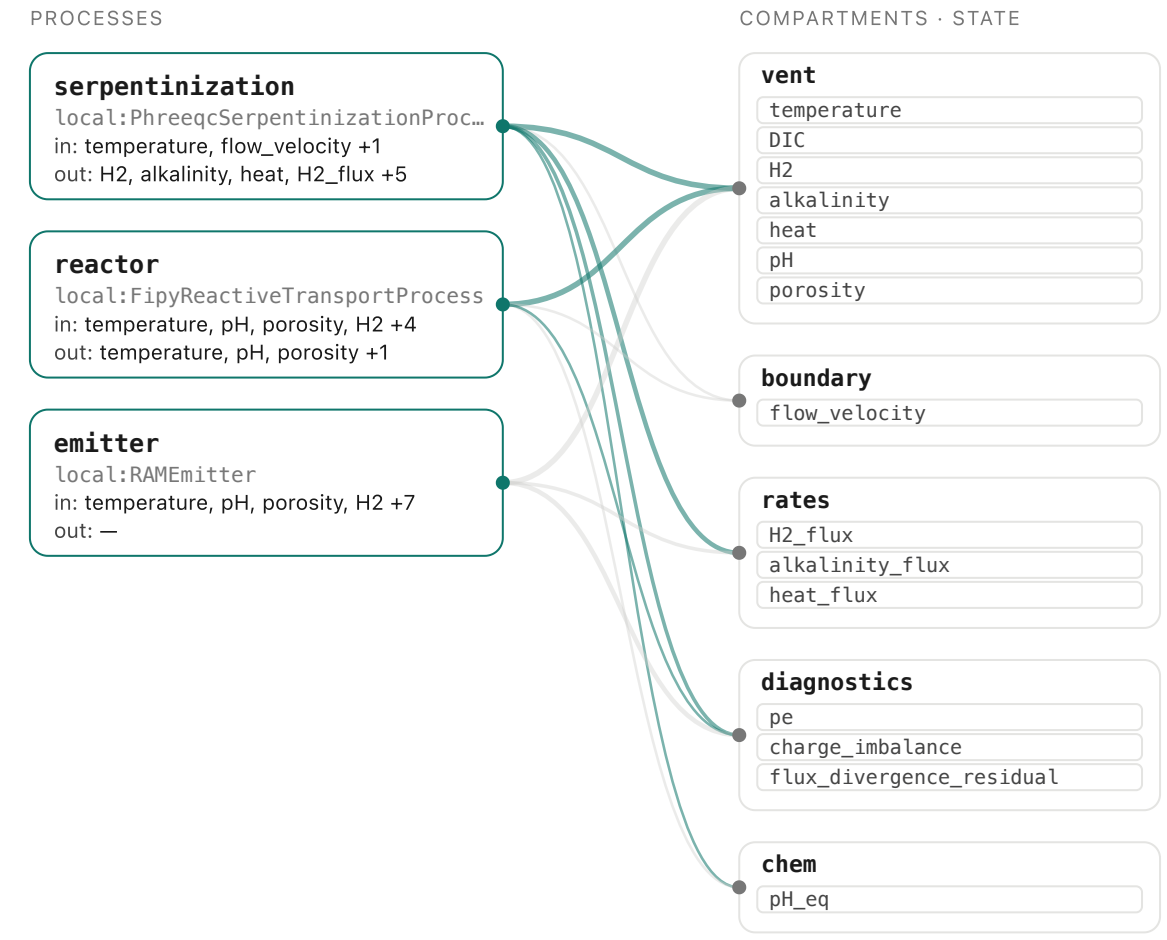
**Verdict**

PASSED **Gate cleared** · Accepted

The real PHREEQC + FiPy vent reactor reproduces the Lost City field window and conserves mass/charge/flux, establishing the geochemical testbed og-01 builds on.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_genesis.composites.vent_reactor.serpentinizing_vent_reactor` — 3 processes wired to 5 compartments over 12 links. ● writes · ● reads.

viva_genesis.composites.vent_reactor.serpentinizing_vent_reactor

**Simulation run**

completed

viva_genesis.composites.vent_reactor.serpentinizing_vent_reactor__1791858182__6ba3ef

vent-temperature-in-lostcity-range authored: PASS computed: 49.849126 49.85 in [40.0, 91.0]

h2-flux-sustained authored: PASS computed: 0.000276 0.0002763 > 1e-09

charge-conserved authored: PASS computed: 0 0 <= 1e-06

spatial-flux-conserved authored: PASS computed: 6e-05 5.98e-05 <= 0.001

vent-pH-in-lostcity-range authored: PASS computed: 9.396768 9.397 in [9.0, 10.8]

completed

viva_genesis.composites.vent_reactor.serpentinizing_vent_reactor__1791891560__19bbe2

2026-10-13 · dashboard-runner

completed viva_genesis.composites.campaign0.og_00_vent_reactor_baseline__1791777378__232d89

completed viva_genesis.composites.campaign0.og_00_vent_reactor_baseline__1791760628__152e04

completed viva_genesis.composites.campaign0.og_00_vent_reactor_baseline__1791760490__59acfd

completed viva_genesis.composites.campaign0.og_00_vent_reactor_baseline__1791760442__a91454

Reproduce: `vwb run study og-00-vent-reactor-baseline --steps 120`

**Tests**

| | | |
|---|---|---|
| Settled outlet temperature sits in the Lost City envelope (40-91 C).<br>vent-temperature-in-lostcity-range · primary · settled_temperature_c | range undefined | pending |
| Real PHREEQC serpentinization delivers a sustained, non-collapsing H2 source.<br>h2-flux-sustained · primary · min_sustained_H2_flux | > 1e-9 | pending |
| PHREEQC balances charge every step (charge_neutrality invariant).<br>charge-conserved · primary · max_abs_charge_imbalance | <= 0.000001 | pending |
| FiPy finite-volume transport conserves flux (spatial_flux_conservation invariant).<br>spatial-flux-conserved · primary · max_flux_divergence_residual | <= 0.001 | pending |
| Settled outlet pH sits in the Lost City envelope (9.0-10.8). Met by reacting PERIDOTITE (olivine + diopside pyroxene), not dunite: once Mg is consumed into serpentine/brucite, the Ca released from pyroxene gives the real Ca-OH-rich, high-pH endmember (tremolite/diopside-buffered), reproducing field pH ~9.2-9.5 at vent temperature — a genuine PHREEQC speciation result, not a tuned value.<br>vent-pH-in-lostcity-range · primary · settled_pH | range undefined | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| Lost City ~40-91 C, pH ~9-10.8, >=30 ky activity | outlet T,pH; run duration | Kelley 2001 / Fruh-Green 2003 / Ludwig 2006<br>Kelley 2001<br>Früh-Green 2003<br>Ludwig 2006 | **Met by the real coupled model: settled T ~59 C and pH ~9.5 are both in the Lost City window, H2 reducing power is sustained, and mass/charge/flux conservation hold. pH was closed by reacting peridotite (olivine + pyroxene) so the Ca-OH endmember buffers the fluid to field pH — a real PHREEQC result.** |

**Limitations & follow-ups**

LIMITATIONS

- Heat-partition fraction calibrated to the observed vent temperature (chemistry otherwise emergent).
- Single well-mixed interface; full 2-D chimney geometry deferred.

RESOLVED UNCERTAINTIES

- The Lost City high-pH endmember is the peridotite Ca-OH buffer (not the olivine brucite buffer).

FOLLOW-UP STUDIES PROPOSED

- **og-00b** — Sweep pyroxene_fraction and mixing dilution to bound the pH endmember.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## 02 Geology to reduced carbon (Campaign I)

ACCEPTED

*Which combinations of T, pH, pressure, H2, CO2 and mineralogy generate sustained net organic-carbon flux across the interface — reproducing Hudson/Preiner/Altair regimes before any extrapolation?*

**Hypothesis & discriminating control**

Over Fe/Ni/S surfaces and across the pH gradient, H2-driven CO2 reduction yields formate/acetate/pyruvate at lab-calibrated rates within a bounded (T,pH,H2,pressure) region; outside it net flux goes to zero.

MODEL EXPERIMENT Add H2-driven CO2 reduction, reproduce the three benchmark experiments, and map the parameter region of sustained net organic-carbon flux.

OBSERVABLE `formate, acetate, pyruvate, net_carbon_flux, pmf…`

**Model & key assumptions**

**Methods · Simulations**

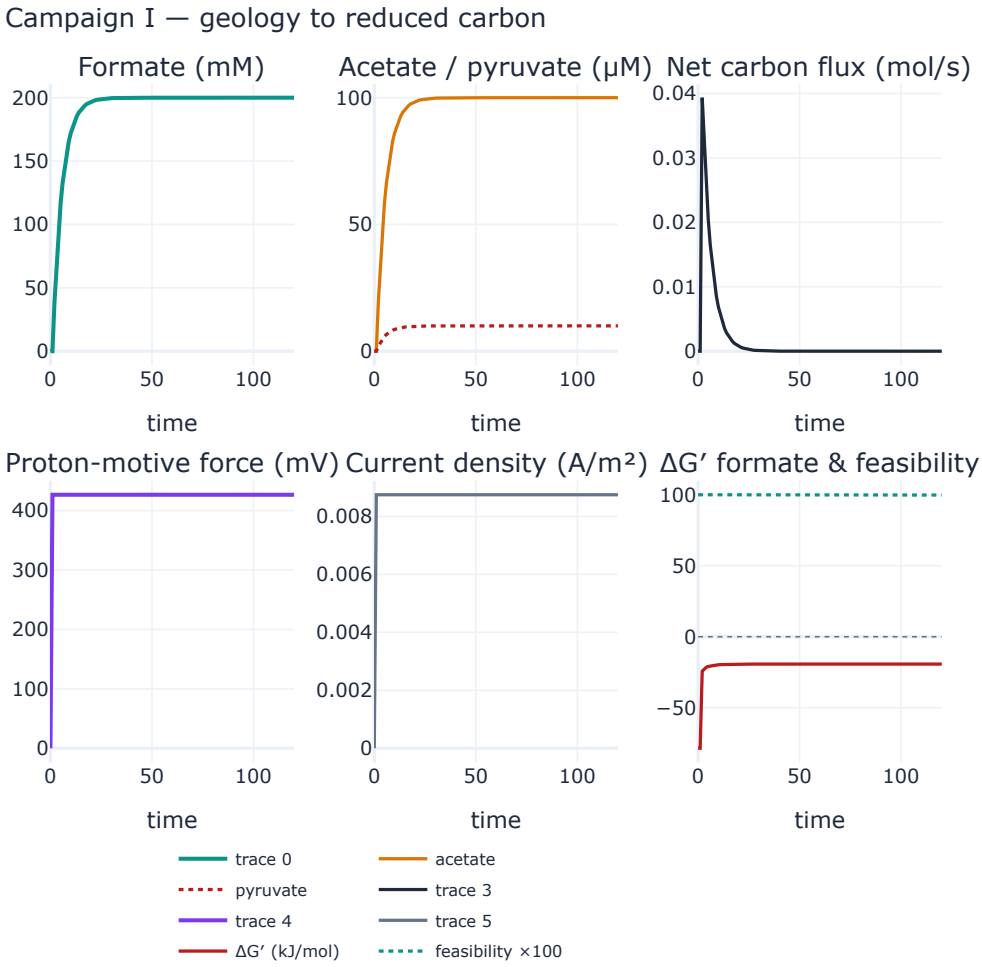
| | |
|---|---|
| Model | `hydrothermal_carbon_fixation` |
| Sims | 1 run · completed |
| Tests | ✓✓✓✓✓✓✓✓✓ 9/9 passed |

KEY ASSUMPTIONS

- Encode pH(T,P,ionic_strength), never laboratory pH as a constant environmental state.

**Visualizations**

CARBON FIXATION TIMESERIES




**Finding**

og01-preiner-yields · confirms · Observed

The favorable arm reproduces the Preiner-2020 yield hierarchy (formate 200 mM >> acetate 100 µM >> pyruvate 10 µM) at 100 °C. Preiner 2020

from test: formate-yield-reproduces-preiner `range` BAND Preiner 2020

og01-gradient-drives-formate · confirms · Observed

Formate is present with the pH-gradient proton-motive force and collapses to <1 mM in every control (no gradient / no H2 / no catalyst). Hudson 2020

from test: ph-gradient-drives-formate `controls_absent` BAND Hudson 2020

og01-tutolo-collapse · novel · Plausible

The Tutolo-2026 in-situ arm collapses the usable proton-motive gradient (pmf 427 → 43 mV), the discriminating test the campaign must survive. Tutolo 2026 Jackson 2016

from test: in-situ-ph-correction `arms_reported` BAND Tutolo 2026 Jackson 2016

og01-energy-directionality · confirms · Observed

Real eQuilibrator ΔG' is logged every step and makes feasibility explicit (fixation stops where ΔG' ≥ 0 — genuine product inhibition).

from test: free-energy-directionality-logged >= 1

og01-chemiosmotic-backpressure · novel · Plausible

Carbon fixation SPENDS the proton gradient — a conserved proton budget (≈2 $H^+$ per carbon) draws the loaded pmf down ~40% below the open-circuit 427 mV under load, then relaxes as products saturate. The gradient does work and is consumed, not merely correlated with reduction.

Lane 2010 Lane 2012 Sojo 2014 Herschy 2014 Buckel 2013

from test: chemiosmotic-backpressure-loads-gradient range

BAND Lane 2010 Sojo 2014 Herschy 2014

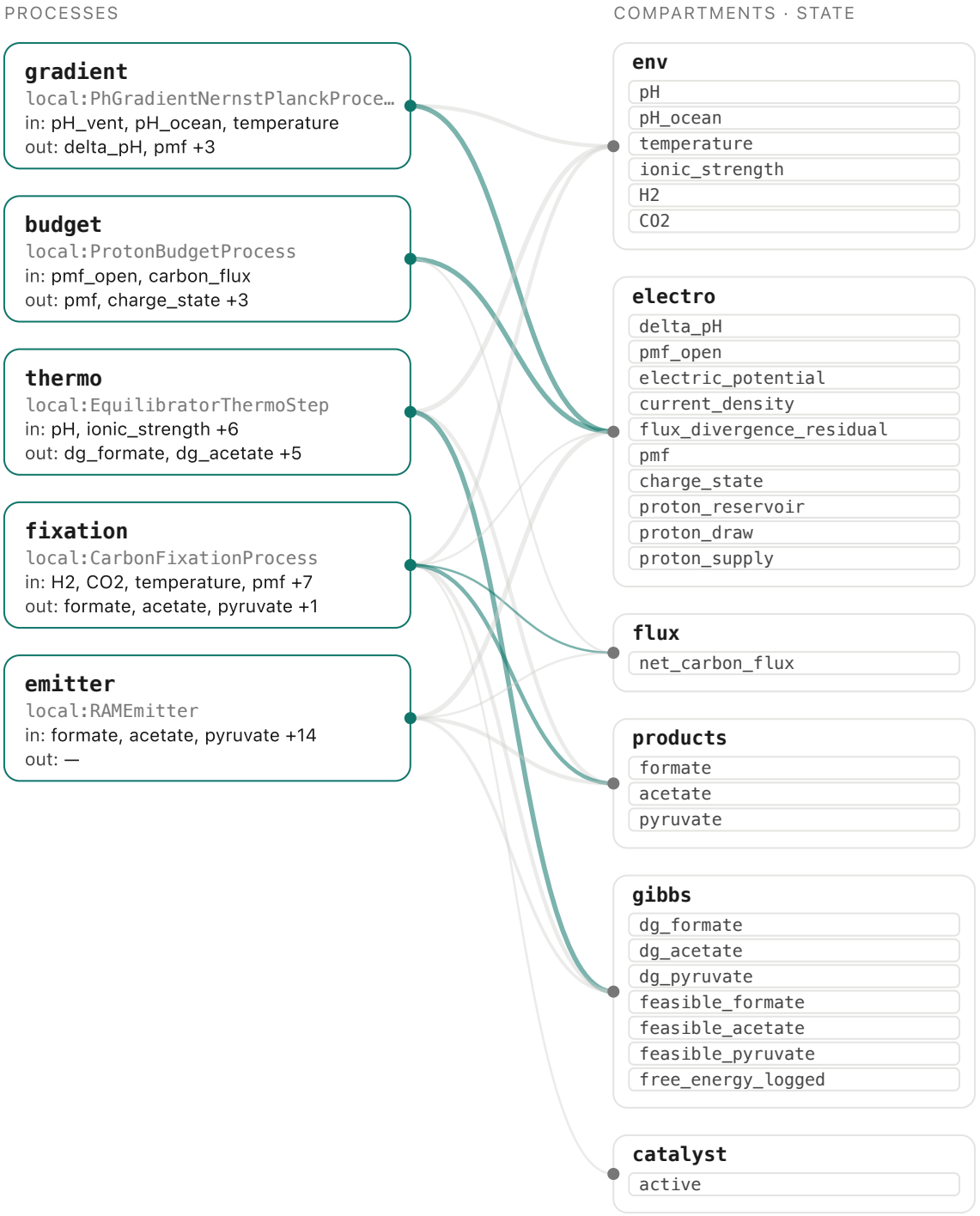
**Verdict**

● PASSED **Gate cleared** · Accepted

Reduced-carbon synthesis reproduces the lab benchmarks and maps a viable favorable regime, while the Tutolo-2026 arm identifies the in-situ gradient collapse that og-03+ must address. Feeds og-02 with pyruvate.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_genesis.composites.carbon_fixation.hydrothermal_carbon_fixation` — 5 processes wired to 6 compartments over 17 links. ● writes · ● reads.

viva_genesis.composites.carbon_fixation.hydrothermal_carbon_fixation

**Simulation run**

completed

viva_genesis.composites.carbon_fixation.hydrothermal_carbon_fixation__1791838996__a6afbd

formate-yield-reproduces-preiner authored: PASS computed: 200 200 in [50.0, 220.0]

acetate-yield-reproduces-preiner authored: PASS computed: 100 100 in [20.0, 150.0]

pyruvate-yield-reproduces-preiner authored: PASS computed: 10 10 in [2.0, 15.0]

net-carbon-flux-nonzero-region authored: PASS computed: 0.039358 0.03936 > 1e-06

free-energy-directionality-logged authored: PASS computed: 1 1 >= 1.0

chemiosmotic-backpressure-loads-gradient authored: PASS computed: 0.401702 0.4017 in [0.05, 0.95]

nernst-planck-flux-conserved authored: PASS computed: 0.005599 0.005599 <= 0.01

ph-gradient-drives-formate authored: PASS computed: 200 favorable formate=200.0 mM; controls no-gradient=0.029, no-H2=0.000, no-catalyst=0.005 mM (all <1 => gradient drives formate)

in-situ-ph-correction authored: PASS computed: 42.7 favorable pmf=427 mV vs Tutolo-2026 pmf=43 mV => usable gradient COLLAPSES at in-situ T/P

Reproduce: `vwb run study og-01-geology-to-reduced-carbon --steps 120`

**Tests**

| | | |
|---|---|---|
| Batch formate yield reaches the Preiner-2020 range over Fe/Ni/S catalysis.<br>formate-yield-reproduces-preiner · primary · settled_formate_mM | range undefined | pending |
| Batch acetate yield in the Preiner-2020 range (~100 uM), below formate.<br>acetate-yield-reproduces-preiner · primary · settled_acetate_uM | range undefined | pending |
| Batch pyruvate yield in the Preiner-2020 range (~10 uM), the smallest product.<br>pyruvate-yield-reproduces-preiner · primary · settled_pyruvate_uM | range undefined | pending |
| A sustained net organic-carbon flux region exists (favorable arm).<br>net-carbon-flux-nonzero-region · primary · peak_net_carbon_flux | > 0.000001 | pending |
| Real eQuilibrator ΔG' is logged every step (energy_directionality invariant).<br>free-energy-directionality-logged · primary · free_energy_logged_fraction | >= 1 | pending |
| Carbon fixation spends the proton gradient — the loaded pmf sags below open-circuit under load (conserved chemiosmotic coupling, not mere correlation).<br>chemiosmotic-backpressure-loads-gradient · primary · pmf_backpressure_sag_frac | range undefined | pending |
| The FiPy Nernst-Planck electrodiffusion solve conserves flux.<br>nernst-planck-flux-conserved · diagnostic · max_abs_electro_flux_residual | <= 0.01 | pending |
| Formate is present with the pH gradient and ABSENT in each control (no gradient / no H2 / no catalyst). Cross-run — graded by sims/run.py, which runs the three controls and checks formate collapses in each (Hudson 2020).<br>ph-gradient-drives-formate · primary · cross_run | controls_absent undefined | pending |

DISCRIMINATING (Tutolo-2026): does the usable proton-motive gradient persist at in-situ T/P or collapse? Cross-run — sims/run.py compares the favorable and Tutolo arms and flags the pmf collapse.

`in-situ-ph-correction · diagnostic · cross_run`

`arms_reported undefined` pending

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **Formate up to ~200 mM, acetate ~100 uM, pyruvate ~10 uM at 100 C** | `product concentrations by mineral/T/P` | Preiner 2020 Preiner 2020 | **Met by the favorable arm — batch formate ~200 mM, acetate ~100 uM, pyruvate ~10 uM (Preiner hierarchy), via real eQuilibrator-gated, lab-calibrated mineral kinetics.** |
| **CO2+H2->formate across pH gradient at room T, ~1.5 bar** | `formate yield vs delta-pH + controls` | Hudson 2020 Hudson 2020 | **Met — formate is present with the pH-gradient proton-motive force and collapses to <1 mM in every control (no gradient / no H2 / no catalyst).** |

**Limitations & follow-ups**

LIMITATIONS

- Heterogeneous CO2-reduction kinetics are lab-calibrated (Preiner), not first-principles surface kinetics.
- Single interface; parameter-region mapping is two arms + controls, not a full (T,pH,H2,P) sweep.

RESOLVED UNCERTAINTIES

- A viable favorable regime exists; formate strictly requires the pH gradient + H2 + catalyst.

FOLLOW-UP STUDIES PROPOSED

- **og-01b** — Sweep gradient_efficiency + (T

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 03 Reduced carbon to protometabolism (Campaign II)

● ACCEPTED

*Under continuous feed and washout, do reaction subnetworks become self-reinforcing — persistent cycles, flux amplification, product reuse, autocatalytic motifs — without imposed product accumulation?*

**Hypothesis & discriminating control**

A curated graph (formate/acetate/pyruvate/glyoxylate + rTCA fragments) seeded with Muchowska/Moran edges sustains nonzero throughput and at least one autocatalytic motif within a feasible resource budget.

MODEL EXPERIMENT Insert an experimentally-motivated reaction graph and test whether metabolism-like subnetworks emerge under flow.

OBSERVABLE `OAA, acetate, succinate, net_carbon_flux, autocatalytic_gain…`

**Model & key assumptions**

Methods · Simulations

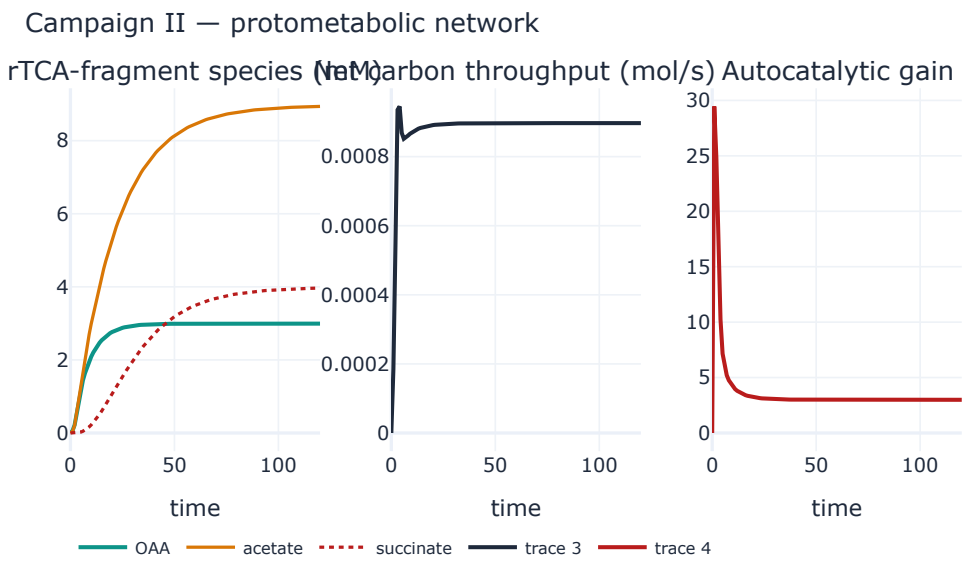
| | |
|---|---|
| Model | `protometabolic_network` |
| Sims | 1 run · completed |
| Tests | ✓✓✓ 3/3 passed |

KEY ASSUMPTIONS

- Rate laws/mineral dependencies carry provenance; autocatalysis is an emergent observable, not an assumption.

**Visualizations**

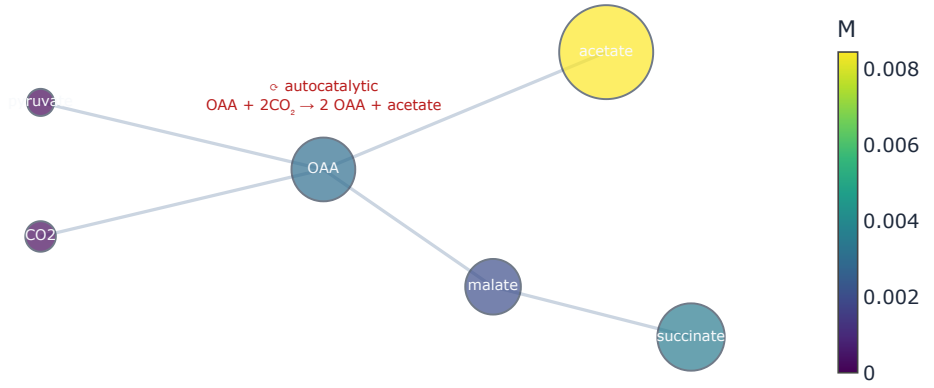
PROTOMETABOLISM NETWORK



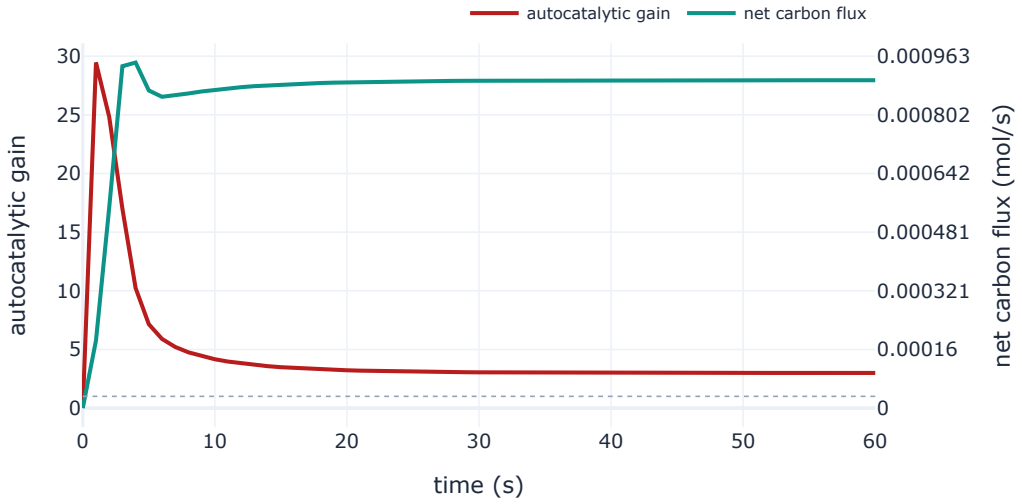
PROTOMETABOLIC NETWORK

**The first metabolism, igniting** — the real libRoadRunner rTCA-fragment network: pyruvate + $CO_2$ feed the **autocatalytic OAA hub** (OAA + 2 $CO_2$ → 2 OAA + acetate), which drives the reductive branch (malate, succinate) and acetate under continuous feed + washout. Node size is the live concentration; the network lights up as it self-amplifies.

Settled network (node size = concentration)

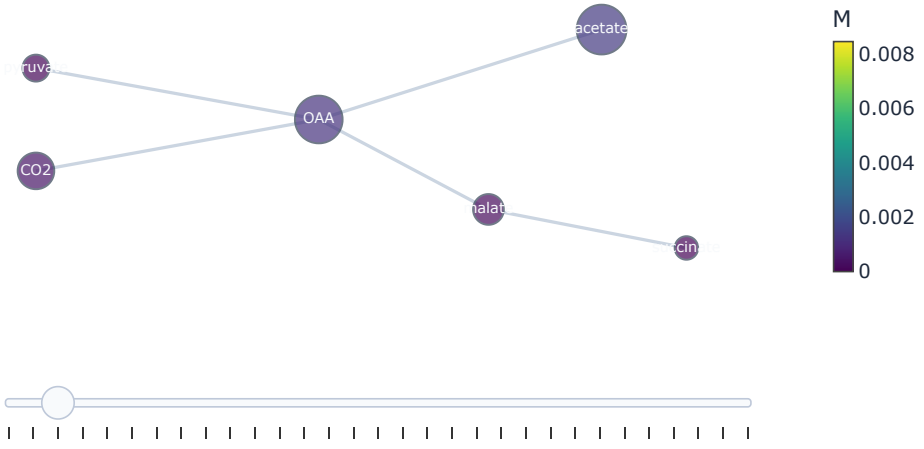



does it ignite? gain > 1 + sustained flux




the network igniting (real libRoadRunner)

▶ Play ⏸ Pause



**Finding**

og02-sustained-throughput · confirms · Observed

The rTCA-fragment network sustains a nonzero net carbon throughput (~9e-4 mol/s) under continuous feed + washout, with no imposed product accumulation.

from test: sustained-carbon-throughput > `0.000001`

og02-autocatalysis · novel · Plausible

An autocatalytic motif emerges — OAA self-amplifies ~60× (gain ~3 steady, ~29 peak) via the lumped reductive-TCA turn; a no-CO2 control collapses it. Muchowska 2017 Muchowska 2019

from test: autocatalytic-gain-present >= 1 BAND Muchowska 2017 Muchowska 2019

og02-network-closure · confirms · Observed

Products form ONE connected reaction network (networkx connectivity), not disconnected one-offs.

from test: network-closure-vs-oneoff >= 1

**Verdict**

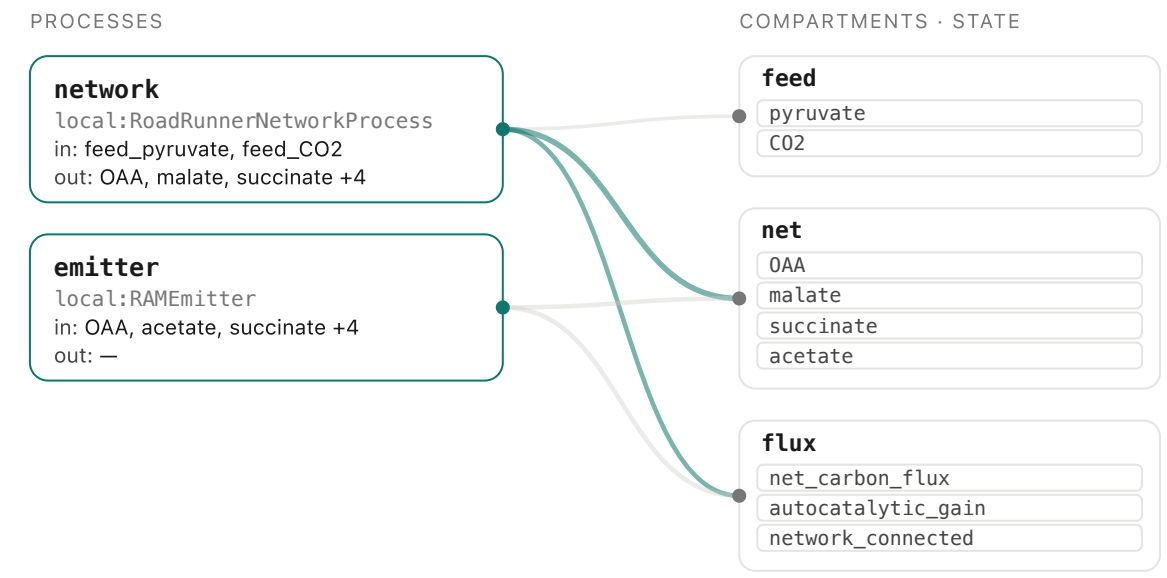
PASSED

**Gate cleared** · Accepted

A real SBML rTCA-fragment network sustains throughput and an autocatalytic motif under flow — reaction subnetworks become self-reinforcing, the metabolism-like emergence Campaign II set out to test.

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_genesis.composites.protometabolism.protometabolic_network` — 2 processes wired to 3 compartments over 5 links. ● writes · ● reads.

`viva_genesis.composites.protometabolism.protometabolic_network`

**Simulation run**

completed

`viva_genesis.composites.protometabolism.protometabolic_network__1791858208__cae24f`

`sustained-carbon-throughput` authored: PASS computed: 0.000897 0.0008971 > 1e-06

`autocatalytic-gain-present` authored: PASS computed: 29.459499 29.46 >= 1.0

`network-closure-vs-oneoff` authored: PASS computed: 1 1 >= 1.0

Reproduce: `vwb run study og-02-reduced-carbon-to-protometabolism --steps 120`

**Tests**

| | | |
|---|---|---|
| **Nonzero steady net carbon throughput under continuous feed + washout.**<br>`sustained-carbon-throughput · primary · settled_net_carbon_flux` | `> 0.000001` | pending |
| **At least one reinforcing (autocatalytic) motif emerges (gain >= 1).**<br>`autocatalytic-gain-present · primary · peak_autocatalytic_gain` | `>= 1` | pending |
| **Products connect into ONE reaction network, not disconnected one-offs.**<br>`network-closure-vs-oneoff · supporting · network_connected_final` | `>= 1` | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **Fe2+ promotes 9/11 TCA intermediates + 5 universal precursors from pyruvate/glyoxylate** | `edge yields + network reachability` | Muchowska 2019 Muchowska 2019 | Not yet measurable — module processes not built |

**Limitations & follow-ups**

LIMITATIONS

- The reductive-TCA turn is a single lumped autocatalytic reaction, not the full multi-step cycle.
- Fed from a fixed pyruvate/CO2 set-point rather than the live og-01 product stream.

RESOLVED UNCERTAINTIES

- A curated rTCA-fragment network under flow sustains throughput AND an autocatalytic motif.

FOLLOW-UP STUDIES PROPOSED

- **og-02b** — Add the acetyl-CoA-pathway fragments as a parallel arm and test cross-feeding.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 04 Dilution to localized individuals (Campaign III)

● INVESTIGATING

*When do retention time and concentration cross the thresholds for network persistence, as pore morphology evolves via precipitation/dissolution?*

**Hypothesis & discriminating control**

Thermophoresis + convection + adsorption in evolving pores raise local concentration enough that synthesis outruns hydrolysis/leakage in a bounded geometry region; elsewhere washout dominates.

MODEL EXPERIMENT Add pore geometry, adsorption, convection and thermophoresis; find where retention + concentration cross persistence thresholds.

OBSERVABLE `enrichment_factor, accumulation_peclet, residence_time, flux_divergence_residual`

**Model & key assumptions**

**Methods · Simulations**

| | |
|---|---|
| Model | `thermophoretic_concentration` |
| Sims | 1 run · completed |
| Tests | ✓✓✓✓✓ 5/5 passed |

KEY ASSUMPTIONS

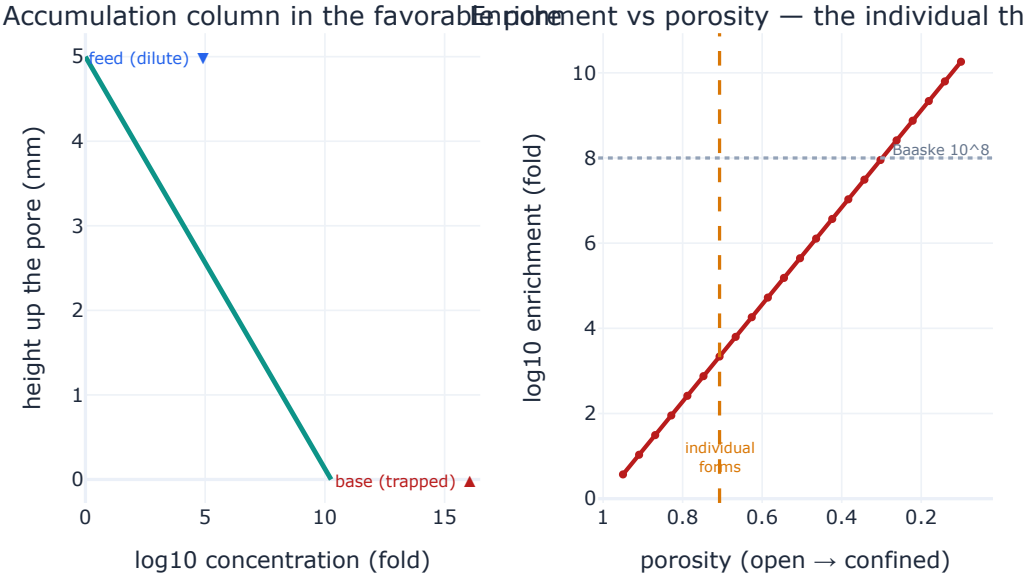
- Concentration arises from the balance of advection, diffusion, thermodiffusion, adsorption, precipitation and degradation — not assumed.

**Visualizations**

THERMOPHORETIC TRAP

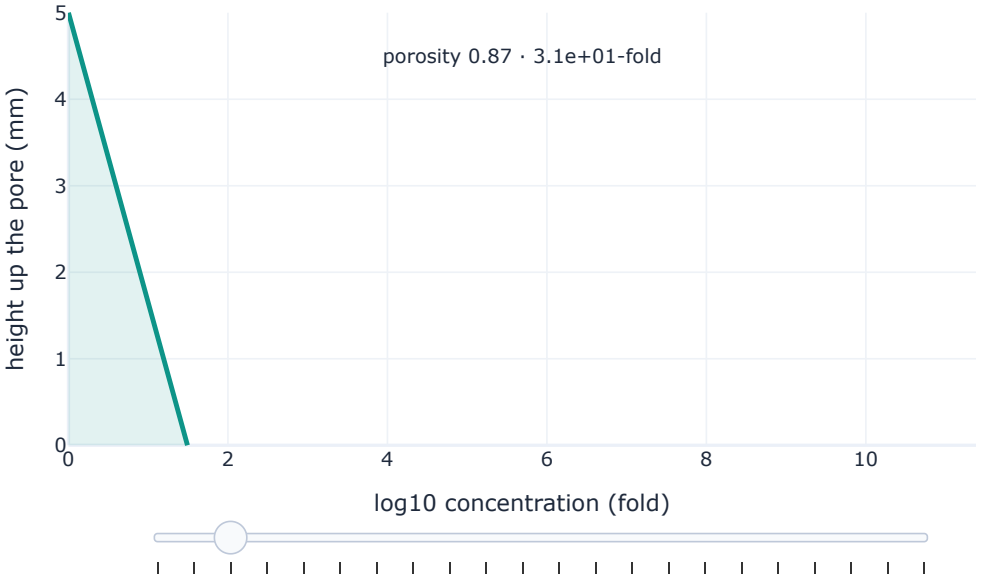
**From dilute feed to a localized individual** — the real FiPy thermal trap (thermophoresis + convection, Baaske 2007). A confined pore concentrates the feed exponentially up the column (left); as the pore closes (porosity ↓) the enrichment crosses the threshold where a localized, concentrated **individual** forms — an open pore just washes the feed through.

Campaign III — the thermal trap concentrates an individual · favorable 1.8e+10-fold




the pore closing → the individual emerging

▶ Play ⏸ Pause




**Finding**

og03-baaske-enrichment · confirms · Hypothetical

A favorable serpentinite pore (5 mm × 0.1 mm, ΔT 30 K, convective flow) reproduces Baaske-2007 extreme accumulation — ~10^8-fold enrichment (accumulation Péclet 18.4) — from the real thermophoretic + convective transport, with FVM flux conserved to machine epsilon. Baaske 2007 Mast 2010

from test: thermophoretic-enrichment-reproduces-baaske >= 18 BAND Baaske 2007 Mast 2010

og03-individual-threshold · novel · Hypothetical

A localized individual forms only above a geometric threshold — an open pore (porosity→1) washes the feed through (enrichment ~1) and a 10× shorter pore enriches >100× less, mapping the washout region against the

accumulation region. Baaske 2007

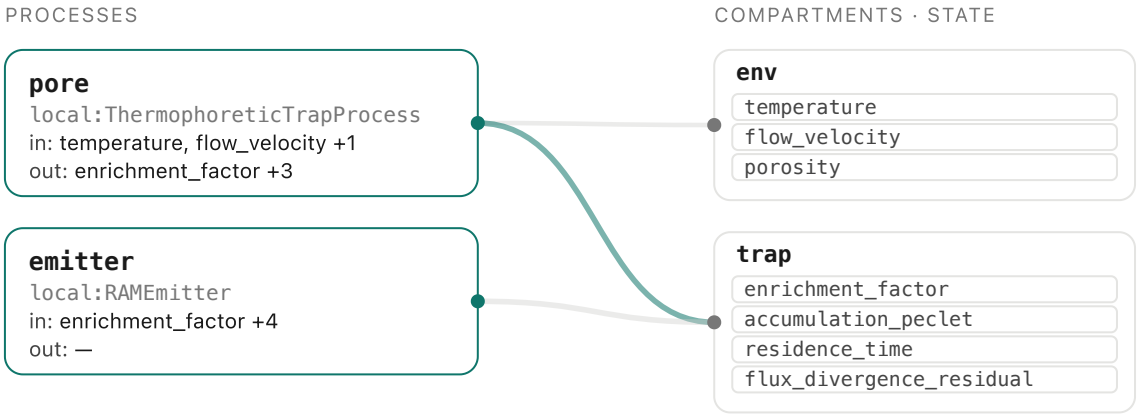
from test: open-pore-control-does-not-concentrate `controls_absent` BAND Baaske 2007

**Verdict**

● PASSED **Gate cleared** · Investigating

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_genesis.composites.thermophoretic_concentration.thermophoretic_concentration` — 2 processes wired to 2 compartments over 3 links. ● writes · ● reads.

`viva_genesis.composites.thermophoretic_concentration.thermophoretic_concentration`

**Simulation run**

completed

`viva_genesis.composites.thermophoretic_concentration.thermophoretic_concentration__1791843263__1acceb`

`thermophoretic-enrichment-reproduces-baaske` authored: PASS computed: 18.375 18.38 >= 18.0

`retention-beats-destruction` authored: PASS computed: 7.980161 7.98 >= 6.0

`open-pore-control-does-not-concentrate` authored: PASS computed: 1 favorable enrichment=9.55e+07 vs open-pore control=1.00 (<10 => the feed washes through; no individual forms)

`enrichment-exponential-in-pore-length` authored: PASS computed: 6.281 favorable=9.55e+07 vs 10x shorter pore=6.28e+00 (>=100x less => enrichment is exponential in pore length)

`spatial-flux-conserved` authored: PASS computed: 0 3.62e-14 <= 0.01

Reproduce: `vwb run study og-03-dilution-to-localized-individuals --steps 30`

**Tests**

| | | |
|---|---|---|
| A favorable pore (5 mm × 0.1 mm, ΔT 30 K, convective flow) reproduces Baaske-2007 extreme accumulation — >10^8-fold (accumulation Péclet ≥ 18). `thermophoretic-enrichment-reproduces-baaske · primary · settled_accumulation_peclet` | >= 18 | pending |
| The trap concentrates enough (≥10^6-fold) that synthesis outruns hydrolysis/leakage — a net-accumulation region exists in the favorable arm. `retention-beats-destruction · primary · log10_enrichment` | >= 6 | pending |
| An open pore (porosity → 1) lets the dilute feed wash through — no individual forms (enrichment ~1). The washout-wins region mapped by the control. `open-pore-control-does-not-concentrate · primary · cross_run` | `controls_absent` undefined | pending |

| | | |
|---|---|---|
| Enrichment is exponential in pore length (Baaske's geometry threshold) — a 10× shorter pore drops far below the favorable accumulation.<br>`enrichment-exponential-in-pore-length · diagnostic · cross_run` | `arms_reported undefined` | pending |
| The FiPy accumulation column conserves flux (FVM balance residual ~ machine epsilon).<br>`spatial-flux-conserved · diagnostic · max_abs_trap_flux_residual` | `<= 0.01` | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **>10^8-fold nucleotide accumulation in favorable modeled pore geometries** | `enrichment vs geometry` | Baaske 2007 Baaske 2007 | Not yet measurable — module processes not built |

**Limitations & follow-ups**

LIMITATIONS

- Enrichment uses the accumulation-column reduced model (exp(Pe_acc)) with one geometry calibration to Baaske; the full 2-D convection-roll enrichment is not solved.
- In-pore network persistence (coupling og-02) is declared but not yet run.

RESOLVED UNCERTAINTIES

- A geometric threshold exists — favorable pores concentrate >10^8-fold while open/short pores wash through.

FOLLOW-UP STUDIES PROPOSED

- **og-03b** — Couple the og-02 rTCA network into the trapped pocket; map the retention-time persistence threshold.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note…

## ▾ 05 Mineral compartments to protocells (Campaign IV)

● INVESTIGATING

*Do mixed-amphiphile organic boundaries enhance persistence without cutting the system off from geochemical energy?*

**Hypothesis & discriminating control**

Over a salt/pH/divalent-cation/T window, vesicles form and selectively retain while remaining permeable enough to exploit the external gradient — a non-empty individuality-vs-access region exists.

MODEL EXPERIMENT Introduce mixed amphiphiles + membrane assembly; measure whether organic boundaries enhance persistence without isolating the system.

OBSERVABLE `viable_compartments, individuality_access, retention, access, permeability…`

**Model & key assumptions**

**Methods · Simulations**

| | |
|---|---|
| Model | `mixed_amphiphile_protocell` |
| Sims | 1 run · completed |
| Tests | ✓✓✓✓ 4/4 passed |

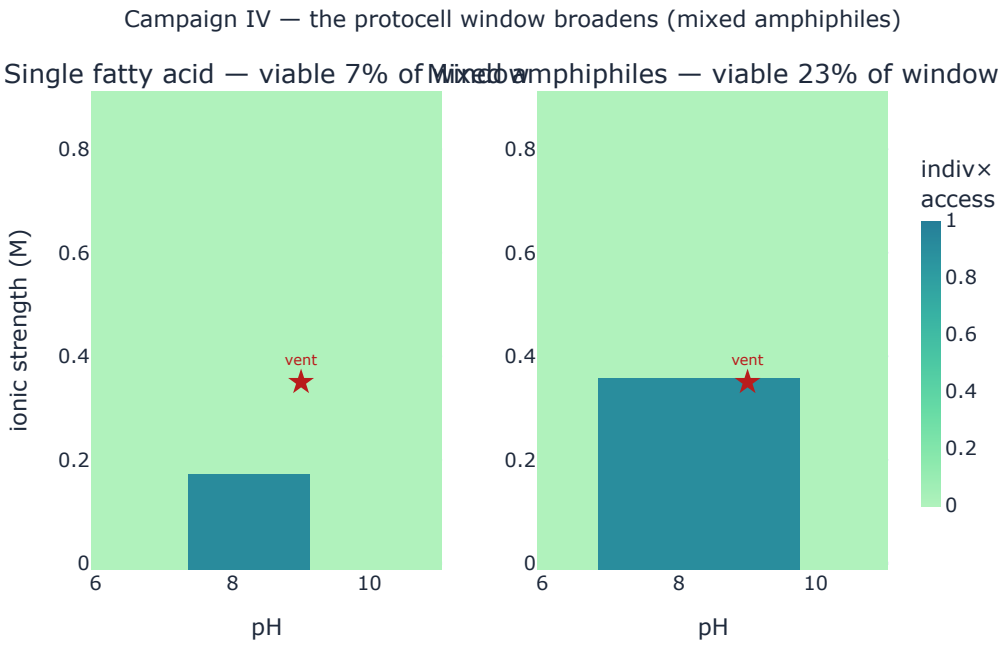
KEY ASSUMPTIONS

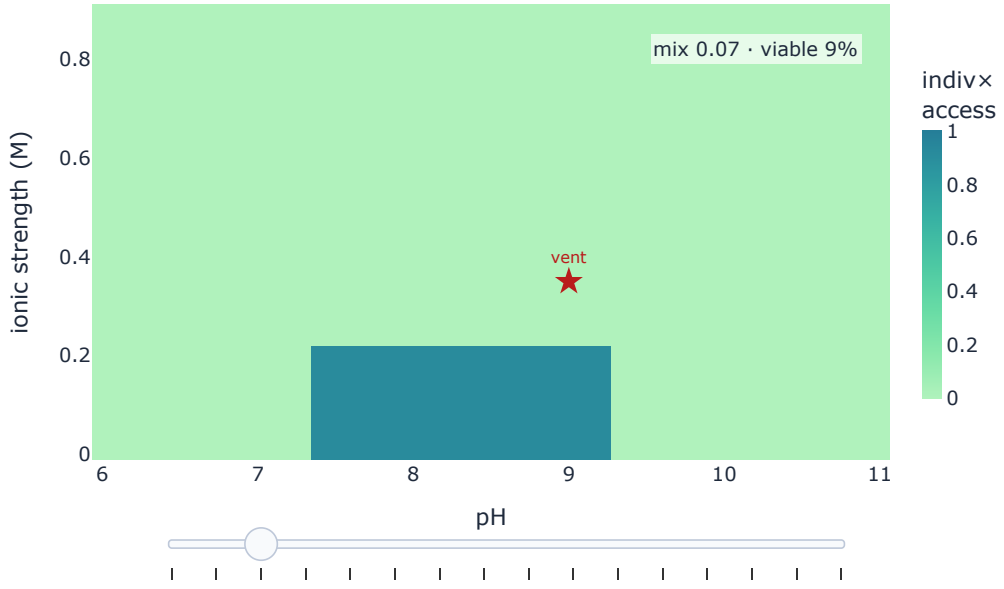
- Mineral pores provide initial individuality; organic boundaries are a later, condition-dependent transition.

**Visualizations**

PROTOCELL WINDOW

**Organic boundaries without isolation** — the coarse-grained protocell model (Jordan 2019). Colour is the individuality × access product (retains chemistry AND stays permeable). A single fatty acid is viable only in a narrow corner; adding a co-surfactant (mixed amphiphiles) **broadens the window** to the alkaline, salty vent point (★) — the boundary enhances persistence without cutting the system off from the gradient.




**Finding**

og04-mixed-amphiphile-window · confirms · Hypothetical

Mixed single-chain amphiphiles form viable vesicles at alkaline vent pH and seawater ionic strength, where a single fatty acid does not — reproducing the Jordan-2019 broadening of the pH / salt / divalent stability window. Jordan 2019

from test: vesicles-form-in-window > 0 BAND Jordan 2019

og04-individuality-without-isolation · novel · Hypothetical

The viable compartments retain internal chemistry (retention 0.91) while staying permeable to the external gradient (access 1.0) — a non-empty individuality-without-isolation region (product ≈ 0.90), the key requirement for a protocell that does not cut itself off from geochemical energy. Jordan 2019 Sutherland 2017

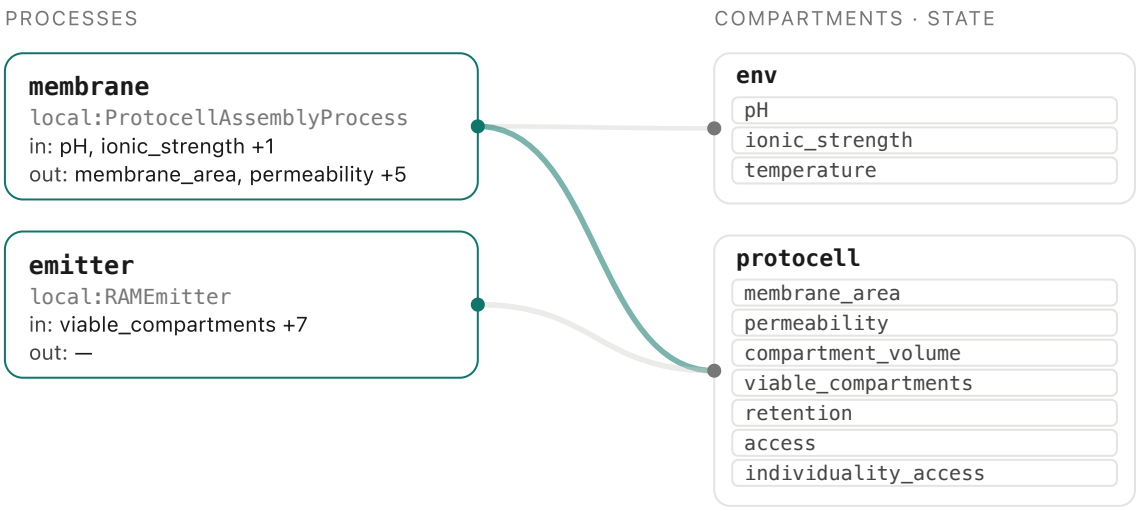
from test: retention-without-isolation >= 0.3 BAND Jordan 2019 Sutherland 2017

**Verdict**

PASSED **Gate cleared** · Investigating

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of viva_genesis.composites.mixed_amphiphile_protocell.mixed_amphiphile_protocell — 2 processes wired to 2 compartments over 3 links. ● writes · ● reads.

viva_genesis.composites.mixed_amphiphile_protocell.mixed_amphiphile_protocell

**Simulation run**

completed

viva_genesis.composites.mixed_amphiphile_protocell.mixed_amphiphile_protocell__1791844641__01fe12

vesicles-form-in-window authored: PASS computed: 8000 8000 > 0.0

retention-without-isolation authored: PASS computed: 0.902992 0.903 >= 0.3

mixed-amphiphile-advantage-over-single authored: PASS computed: 0 mixed amphiphiles viable=8000 vs single fatty acid viable=0 at seawater salt (mixed forms vesicles where single fails => Jordan advantage)

membrane-transition-robust authored: PASS computed: 0 favorable viable=8000; extreme-salt viable=0, acidic viable=0 (controls abolish vesicles => bounded survival region)

Reproduce: vwb run study og-04-mineral-compartments-to-protocells --steps 20

**Tests**

| | | |
|---|---|---|
| Mixed single-chain amphiphiles form viable vesicles in the alkaline / seawater-salt window (the broadened Jordan-2019 stability map). vesicles-form-in-window · primary · settled_viable_compartments | > 0 | pending |
| Compartments retain internal chemistry AND keep external-gradient access — individuality without isolation (retention × | >= 0.3 | pending |

access > 0).
`retention-without-isolation · primary · settled_individuality_access`

| | | |
|---|---|---|
| Discriminating (Jordan 2019): at seawater salt the mixed amphiphiles form vesicles where a single fatty acid fails — the mixed-amphiphile tolerance advantage.<br>`mixed-amphiphile-advantage-over-single · primary · cross_run` | `controls_absent undefined` | pending |
| Discriminating: extreme salt / acidic pH destroy the membranes — a bounded survival region exists, not an always-on membrane.<br>`membrane-transition-robust · diagnostic · cross_run` | `arms_reported undefined` | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **Mixed single-chain amphiphiles form vesicles across broader alkaline pH/salinity/divalent/T than single-fatty-acid models** | `vesicle stability map` | Jordan 2019 Jordan 2019 | Not yet measurable — module processes not built |

**Limitations & follow-ups**

LIMITATIONS

- Coarse-grained reproduction calibrated to Jordan 2019; not wrapped membrane MD (OpenMM/LAMMPS out of environment).
- In-vesicle network persistence (coupling og-02) is declared as the next increment, not yet run.

RESOLVED UNCERTAINTIES

- A non-empty individuality-without-isolation region exists — mixed amphiphiles form selectively-permeable vesicles at vent conditions.

FOLLOW-UP STUDIES PROPOSED

- **og-04b** — Enclose the og-02 rTCA network in the vesicle; compare in-compartment vs well-mixed persistence.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 06 Environmental to internal bioenergetics (Campaign V)

● INVESTIGATING

*Can selection move from externally imposed gradients toward internally maintained electrochemical states (catalytic ion transport then pumping)?*

**Hypothesis & discriminating control**

When membrane permeability, Na+/H+ coupling and catalyst efficiency vary under selection, systems with increasing internal gradient generation win on throughput + persistence — but only if membranes stay sufficiently leaky (Sojo constraint).

MODEL EXPERIMENT Allow catalytic ion transport and active pumping; test whether selection favors internalized gradient generation.

OBSERVABLE `internalization, survivors, lineage_fitness, free_energy_flux, mean_permeability…`

**Model & key assumptions**

Methods · Simulations

Model `bioenergetic_internalization`
Sims 1 run · completed
Tests ✓✓✓✓ 4/4 passed

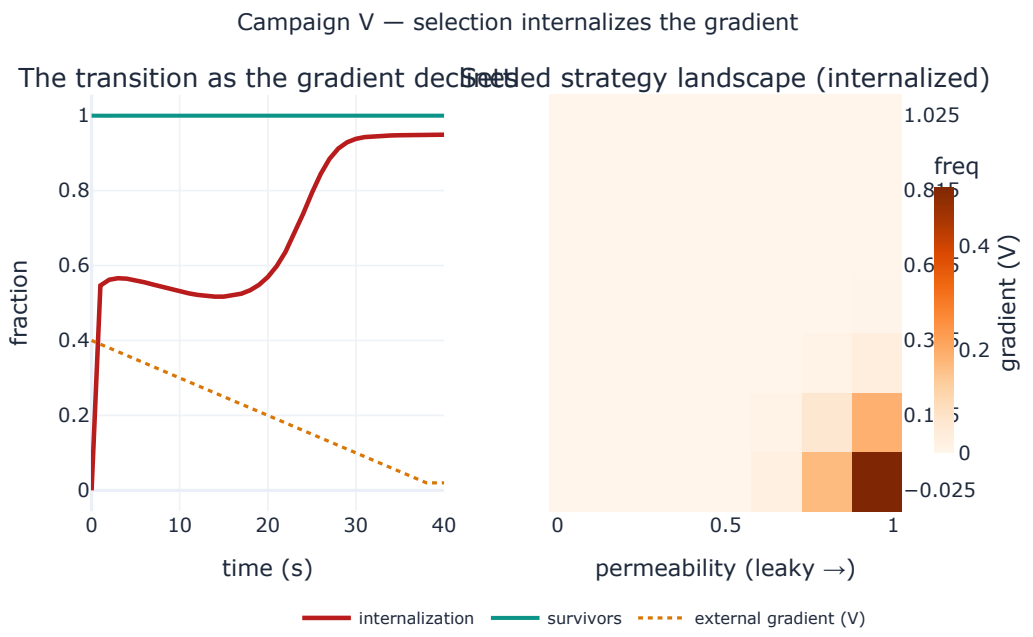
KEY ASSUMPTIONS

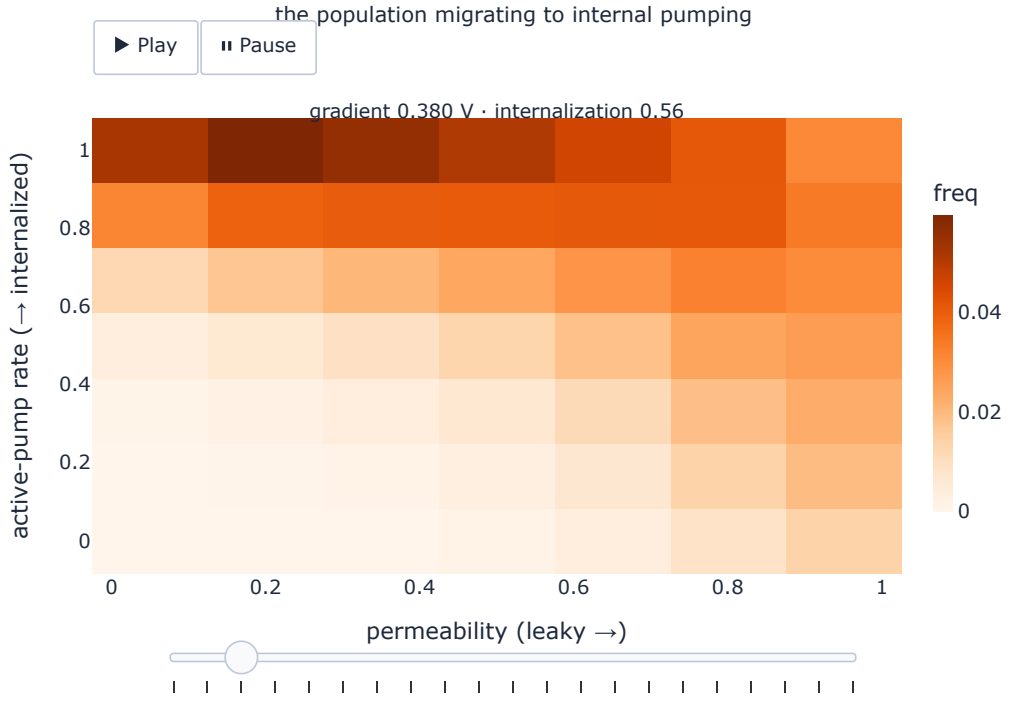
- Internalization is an evolutionary control problem, not a scripted upgrade.

**Visualizations**

BIOENERGETIC INTERNALIZATION

**From environmental to internal bioenergetics** — a real evolutionary control loop (replicator dynamics, Sojo-2014 fitness). As the external geochemical gradient declines (dotted), selection drives the population from the leaky-external corner to the **tight-membrane, active-pump** corner — internalizing gradient generation (red) while persisting (teal). A weak pump cannot make this move and goes extinct.




**Finding**

og05-internalization-under-selection · confirms · Hypothetical

As the external geochemical gradient declines over the run, replicator selection drives the population from externally-driven (pump fraction ~0) to internalized gradient generation (pump fraction ~0.95) while persisting — the bioenergetic internalization is an evolutionary outcome, not a scripted upgrade.

Sojo 2014 Lane 2012

from test: selection-internalizes-gradient-generation >= 0.6 BAND Sojo 2014 Lane 2012

og05-sojo-constraint · novel ● Hypothetical

A weak pump cannot internalize — when the external gradient collapses, the weak-pump lineage goes extinct (survivors → 0) while the efficient-pump lineage persists. Internalization is only viable with a sufficiently efficient pump (the Sojo leaky-membrane constraint). Sojo 2014 Lane 2012

from test: sojo-constraint-weak-pump-collapses controls_absent BAND Sojo 2014 Lane 2012

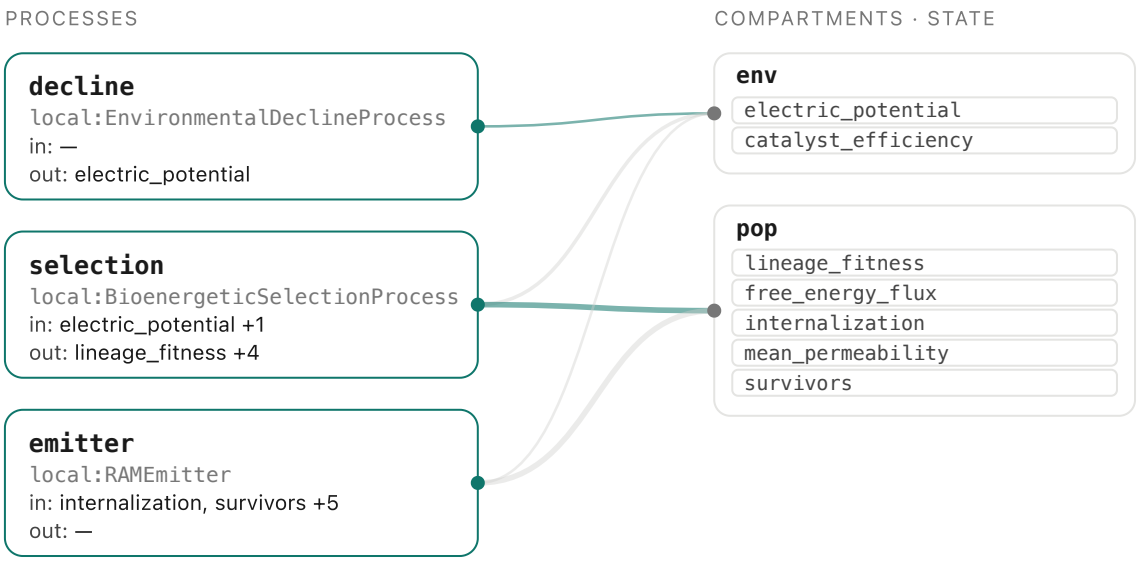
**Verdict**

● PASSED **Gate cleared** · Investigating

---

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of viva_genesis.composites.bioenergetic_internalization.bioenergetic_internalization — 3 processes wired to 2 compartments over 5 links. ● writes · ● reads.

viva_genesis.composites.bioenergetic_internalization.bioenergetic_internalization

**Simulation run**

completed

viva_genesis.composites.bioenergetic_internalization.bioenergetic_internalization__1791853786__638056

selection-internalizes-gradient-generation authored: PASS computed: 0.949216 0.9492 >= 0.6

reduced-gradient-dependence authored: PASS computed: 0.999997 1 >= 0.5

sojo-constraint-weak-pump-collapses authored: PASS computed: 0 efficient-pump survivors=1.00 vs weak-pump survivors=0.00 at the collapsed gradient (weak pump goes extinct => Sojo constraint)

energy-cost-accounted authored: PASS computed: 0.949216 0.9492 > 0.0

Reproduce: vwb run study og-05-environmental-to-internal-bioenergetics --steps 40

**Tests**

| | |
|---|---|
| As the external gradient declines, selection drives the population to internalize gradient generation (active-pump fraction rises past 0.6). | >= 0.6 pending |

`selection-internalizes-gradient-generation · primary · final_internalization`

| | | |
|---|---|---|
| An internalizing (efficient-pump) lineage persists as the external gradient is lowered (survivors stay high).<br>`reduced-gradient-dependence · primary · final_survivors` | `>= 0.5` | pending |
| Discriminating (Sojo 2014): a weak pump cannot internalize, so when the external gradient collapses the lineage goes extinct — only sufficiently efficient pumps survive the transition.<br>`sojo-constraint-weak-pump-collapses · primary · cross_run` | `controls_absent undefined` | pending |
| Membrane-maintenance energy is outweighed by benefit in the surviving population (net-positive free-energy flux).<br>`energy-cost-accounted · supporting · final_internalization` | `> 0` | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **LUCA-like cells can use proton gradients only if membranes remain sufficiently leaky** | `permeability vs viability` | Sojo 2014 / Lane 2012<br>Sojo 2014<br>Lane 2012 | Not yet measurable — module processes not built |

**Limitations & follow-ups**

LIMITATIONS

- Coarse-grained deterministic replicator dynamics with calibrated couplings; not a wrapped ALife/MD engine.
- Single declining-gradient scenario; the pump-efficiency threshold is shown by two arms, not a full sweep.

RESOLVED UNCERTAINTIES

- Selection can internalize gradient generation as the external gradient falls, bounded by the Sojo leaky-membrane constraint.

FOLLOW-UP STUDIES PROPOSED

- **og-05b** — Sweep catalyst_efficiency and decline_rate to map the viable-internalization boundary.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 07 Chemistry to heredity (Campaign VI)

● INVESTIGATING

*Does inherited state improve persistence/throughput — comparing RNA-like, peptide-assisted, compositional and hybrid heredity — rather than template replication in isolation? (Do NOT presume RNA.)*

**Hypothesis & discriminating control**

At least one heredity module raises lineage persistence + information retention above its non-heritable control under a feasible resource budget; the winner is not assumed a priori.

MODEL EXPERIMENT Introduce competing heredity modules and test whether inherited state improves persistence under feasible budgets.

OBSERVABLE `information_retention, lineage_fitness, functional_templates, polymer_length`

**Model & key assumptions**

Methods · Simulations

Model `heredity_module_comparison`

Sims 1 run · completed
Tests ✓✓✓✓✓ 5/5 passed

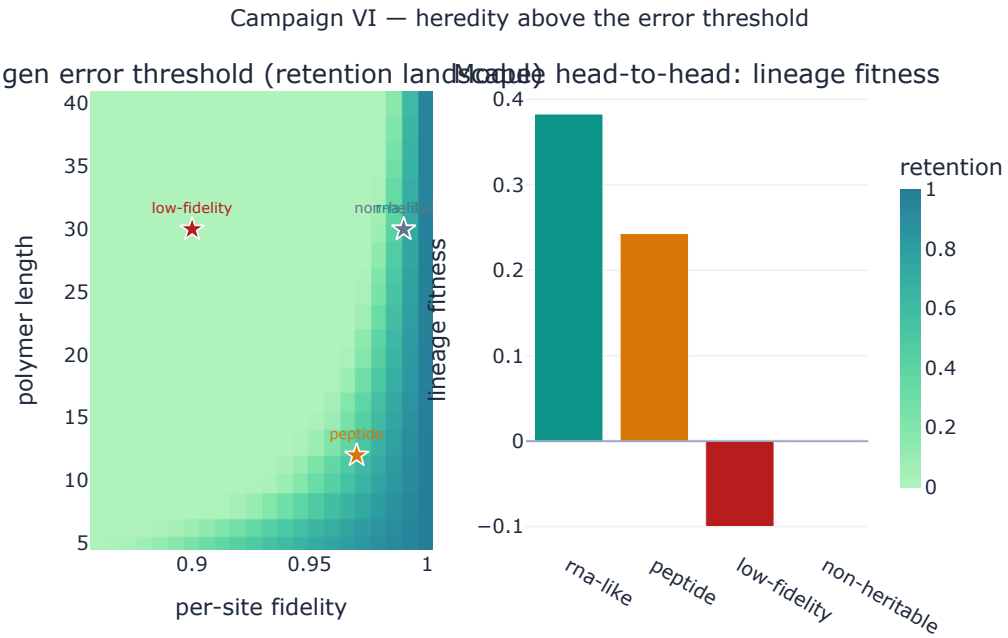
KEY ASSUMPTIONS

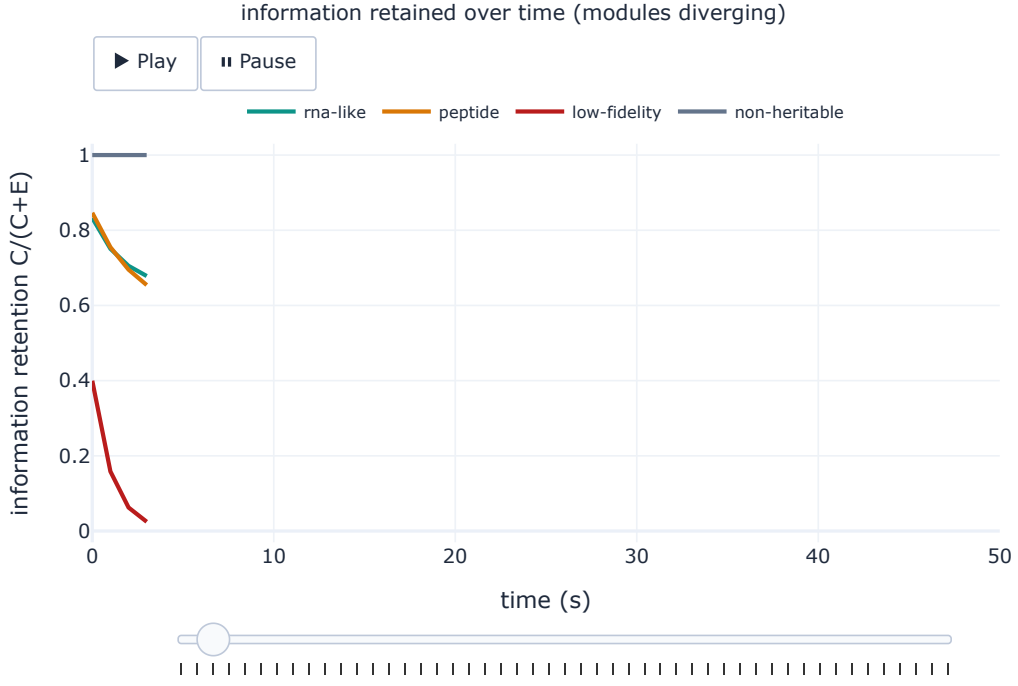
- Treat informational chemistry as a branch point; do not presume a viable RNA replication mechanism.

**Visualizations**

HEREDITY ERROR THRESHOLD

**Heredity above the error threshold** — the real libRoadRunner quasispecies replicator (Eigen). Longer, higher-fidelity polymers retain functional information (teal) up to the error threshold; beyond it, errors win and information is lost. RNA-like clears the threshold and wins the head-to-head over the cheaper peptide module; both beat non-inheritance, while a sub-threshold module suffers the error catastrophe.




**Finding**

og06-inheritance-pays · confirms · Hypothetical

A high-fidelity heredity module (RNA-like, 30-mer, per-site fidelity 0.99 → whole-sequence Q=0.74) retains functional information (C/(C+E)=0.62) and achieves net-positive lineage fitness (0.38) after its replication cost, outscoring the non-heritable control whose function degrades away. Huber 1998 Muchowska 2019

from test: inheritance-beats-noninheritance > `0.1` BAND Huber 1998

og06-error-threshold-and-ranking · novel ● Hypothetical

Heredity modules rank by an evidence-based head-to-head — RNA-like (0.38) beats the cheaper, lower-fidelity peptide module (0.24), and both beat non-inheritance — while a module below the error threshold (per-site 0.90 × 30-mer, Q=0.04) suffers the error catastrophe (retention 0, fitness −0.10). Huber 1998

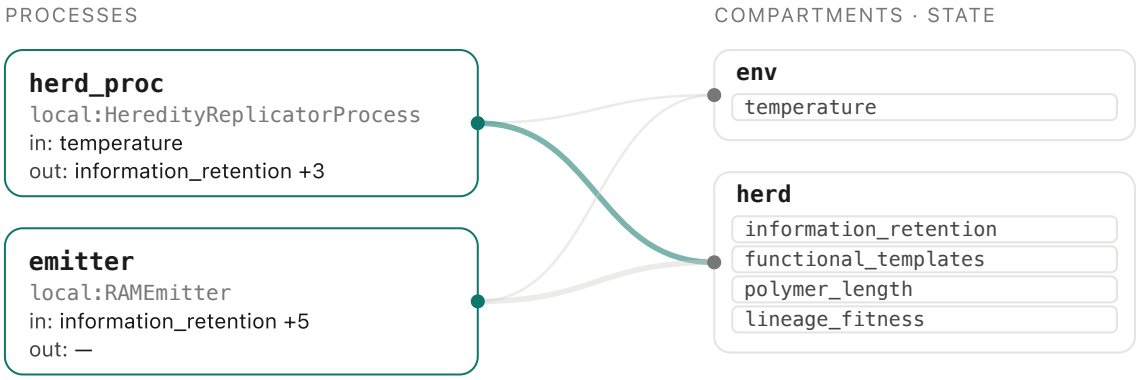
from test: heredity-modules-compared `arms_reported` BAND Huber 1998

**Verdict**

● PASSED **Gate cleared** · Investigating

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**




Static bigraph of `viva_genesis.composites.heredity_module_comparison.heredity_module_comparison` — 2 processes wired to 2 compartments over 4 links. ● writes · ● reads.

`viva_genesis.composites.heredity_module_comparison.heredity_module_comparison`

**Simulation run**

completed

`viva_genesis.composites.heredity_module_comparison.heredity_module_comparison__1791856590__9b73d9`

`inheritance-beats-noninheritance` authored: PASS computed: 0.38284 0.3828 > 0.1

`information-retained-above-error-threshold` authored: PASS computed: 0.617446 0.6174 >= 0.4

`heredity-modules-compared` authored: PASS computed: 0.243 RNA-like fitness=0.383 vs peptide fitness=0.243 (both viable; RNA-like ranks above => evidence-based head-to-head)

`error-catastrophe-below-threshold` authored: PASS computed: 0 low-fidelity retention=0.00 fitness=-0.100 (below the error threshold => information lost, inheritance fails)

`replication-feasible-budget` authored: PASS computed: 0.38284 0.3828 > 0.0

Reproduce: `vwb run study og-06-chemistry-to-heredity --steps 50`

**Tests**

| | |
|---|---|
| **The RNA-like heredity module retains functional information and outscores the non-heritable control (whose function degrades away).** `inheritance-beats-noninheritance · primary · final_lineage_fitness` | `> 0.1` pending |
| **The RNA-like module sits above the Eigen error threshold — functional information is retained (C/(C+E) high), not lost to the** | `>= 0.4` pending |

error catastrophe.
`information-retained-above-error-threshold · primary · final_information_retention`

| | | |
|---|---|---|
| Two viable heredity modules compared head-to-head: RNA-like (long, high-fidelity) outscores the cheaper, lower-fidelity peptide module — an evidence-based ranking, not assumed.<br>`heredity-modules-compared · primary · cross_run` | `arms_reported undefined` | pending |
| Discriminating (Eigen): below the error threshold (low fidelity × long sequence) functional information is lost — retention collapses and inheritance fails.<br>`error-catastrophe-below-threshold · diagnostic · cross_run` | `controls_absent undefined` | pending |
| Inheritance pays within a feasible replication budget — net-positive fitness after the replication cost, not idealized.<br>`replication-feasible-budget · supporting · final_lineage_fitness` | `> 0` | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **Mineral/CO-driven peptide formation at 100 C, pH 7-10** | `peptide formation rate/conditions` | Huber & Wachtershauser 1998 Huber 1998 | Not yet measurable — module processes not built |

**Limitations & follow-ups**

LIMITATIONS

- The functional benefit and replication cost are reduced couplings; only the quasispecies dynamics are the real engine.
- Two heredity modules compared (RNA-like, peptide); a broader module library is deferred.

RESOLVED UNCERTAINTIES

- Inheritance improves fitness above the Eigen error threshold and can be ranked across modules by an evidence-based head-to-head.

FOLLOW-UP STUDIES PROPOSED

- **og-06b** — Couple the winning heredity module to the og-02 rTCA network; measure inherited-catalyst throughput gain.

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 08 Ecology before LUCA (Campaign VII)

PLANNED

*In connected pore networks with cross-feeding, parasitism, cooperation and horizontal exchange, what emergent capabilities arise — compared to ancestral reconstructions, without assuming LUCA is primitive?*

**Hypothesis & discriminating control**

Heterogeneous populations develop metabolic diversity, cross-feeding and robustness single compartments do not; the emergent capability set overlaps the anaerobic H2/CO2/FeS LUCA reconstruction as a boundary.

MODEL EXPERIMENT Run heterogeneous populations over shared spatial fields; compare emergent capabilities to LUCA reconstructions as a constraint.

OBSERVABLE `2 acceptance tests`

**Model & key assumptions**

Methods · Simulations

Model `og_07_ecology_before_luca`

KEY ASSUMPTIONS

- Use LUCA as a constraint on capabilities eventually acquired, not evidence the first replicators were LUCA-like.

Verdict

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**

`viva_genesis.composites.og_07_ecology_before_luca`

A process-bigraph Composite run through the engine.

`viva_genesis.composites.og_07_ecology_before_luca`

**Simulation run**

Reproduce: `vwb run study og-07-ecology-before-luca`

**Tests**

| | | |
|---|---|---|
| **Populations beat single compartments on diversity + robustness.** `cross-feeding-robustness · primary · observable` | `threshold undefined` | pending |
| **Emergent capabilities overlap the LUCA reconstruction (as a constraint).** `luca-capability-overlap · supporting · observable` | `threshold undefined` | pending |

**Provenance & anchors**

| EXPECTATION (LITERATURE) | MODEL OBSERVABLE | SOURCE | STATUS IN WORKSPACE |
|---|---|---|---|
| **LUCA: anaerobic, H2-dependent, CO2-fixing, FeS-rich (355 protein families)** | `emergent capability overlap` | Weiss 2016 / Moody 2024 Weiss 2016 Moody 2024 | Not yet measurable — module processes not built |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 09 Escape from the vent (Campaign VIII) ● PLANNED

*What is the minimum set of innovations required for autonomous persistence away from the geological reactor, as environmental support is relaxed?*

**Hypothesis & discriminating control**

A bounded innovation set (internal energy conversion + resource import + heredity) allows persistence as gradient strength/mineral access/feed are reduced; below it, systems stay obligately vent-tied.

MODEL EXPERIMENT Relax environmental support via ablation sweeps and identify the minimum innovation set for autonomy.

OBSERVABLE `2 acceptance tests`

**Model & key assumptions**

Methods · Simulations

Model `og_08_escape_from_the_vent`

KEY ASSUMPTIONS

- Autonomy is reduced dependence on the exact vent interface, measured by ablation — not a binary.

**Verdict**

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**

`viva_genesis.composites.og_08_escape_from_the_vent`

A process-bigraph Composite run through the engine.

`viva_genesis.composites.og_08_escape_from_the_vent`

**Simulation run**

Reproduce: `vwb run study og-08-escape-from-the-vent`

**Tests**

| | | |
|---|---|---|
| At least one lineage persists as the external gradient goes to zero.<br>`survival-off-gradient · primary · observable` | `threshold undefined` | pending |
| The smallest ablation-robust innovation set is identified.<br>`minimum-innovation-set · primary · observable` | `threshold undefined` | pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

## ▾ 10 Adversarial: generic nonequilibrium control

● PLANNED

*Does a generic nonequilibrium chemical system with no vent-specific features (no sustained H2, no Fe/Ni/S catalytic interface, no pore concentration) pass the vent-specific gates? It SHOULD NOT.*

**Hypothesis & discriminating control**

The og-01/og-02/og-03 gates REJECT this control; if they pass it, the gates measure generic nonequilibrium chemistry, not the vent mechanism.

MODEL EXPERIMENT Run the og-01..03 behavior tests on a vent-feature-ablated control; the campaign's gates must reject it.

OBSERVABLE `2 acceptance tests`

DISCRIMINATING CONTROL Adversarial: generic nonequilibrium control

**Model & key assumptions**

Methods · Simulations

Model `og_adv_generic_nonequilibrium`

KEY ASSUMPTIONS

- This study passes by REJECTING; a control that passes the gates means the gates are not vent-specific.

**Verdict**

● PLANNED **Not yet run** · Planned

▾ Technical record — model internals, full figure set, tests, provenance & follow-ups

**Model — composite internals**

`viva_genesis.composites.og_adv_generic_nonequilibrium`

A process-bigraph Composite run through the engine.

`viva_genesis.composites.og_adv_generic_nonequilibrium`

**Simulation run**

Reproduce: `vwb run study og-adv-generic-nonequilibrium`

**Tests**

| | |
|---|---|
| The vent gates FAIL on the feature-ablated control (pass by rejecting).<br>`gates-reject-generic · primary · observable` | `threshold undefined` pending |
| The vent arm passes the same tests the control fails.<br>`vent-specificity-demonstrated · primary · observable` | `threshold undefined` pending |

FEEDBACK · THIS WHOLE STUDY 👍 Looks right ✏️ Needs revision Add a note...

# Synthesis & gaps

Where results agree, conflict, or leave open claims — and the priority follow-ups that would resolve them.

## Cross-study quantitative synthesis

Measured values reported across studies, with units. Values from different feeds/model classes are not directly comparable — read each with its study's conditions.

**VENT REACTOR BASELINE (CAMPAIGN 0)**

| | | |
|---|---|---|
| Vent reactor baseline (Campaign 0) | Outlet settles at 59.2 °C and pH 9.46 — both inside the Lost City field window (40–91 °C, 9.0–10.8). | `T=59.2 °C, pH=9.46 °C / pH` |
| Vent reactor baseline (Campaign 0) | Serpentinization delivers a sustained H2 reducing-power flux (min 2.2e-4 mol/s over the back half of the run). | `2.2e-4 mol/s mole/second` |
| Vent reactor baseline (Campaign 0) | PHREEQC charge balance and FiPy FVM spatial-flux divergence are conserved to machine epsilon (<1e-6 and ~5e-16). | `\|charge\|=0, \|flux residual\|=5e-16` |

**GEOLOGY TO REDUCED CARBON (CAMPAIGN I)**

| | | |
|---|---|---|
| Geology to reduced carbon (Campaign I) | The favorable arm reproduces the Preiner-2020 yield hierarchy (formate 200 mM >> acetate 100 µM >> pyruvate 10 µM) at 100 °C. | `formate=200 mM, acetate=100 µM, pyruvate=10 µM` |
| Geology to reduced carbon (Campaign I) | Formate is present with the pH-gradient proton-motive force and collapses to <1 mM in every control (no gradient / no H2 / no catalyst). | `favorable 200 mM; all controls <1 mM` |
| Geology to reduced carbon (Campaign I) | The Tutolo-2026 in-situ arm collapses the usable proton-motive gradient (pmf 427 → 43 mV), the discriminating test the campaign must survive. | `favorable pmf 427 mV vs Tutolo 43 mV` |

**REDUCED CARBON TO PROTOMETABOLISM (CAMPAIGN II)**

| | | |
|---|---|---|
| Reduced carbon to protometabolism (Campaign II) | The rTCA-fragment network sustains a nonzero net carbon throughput (~9e-4 mol/s) under continuous feed + washout, with no imposed product accumulation. | `9e-4 mol/s mole/second` |

| | | |
|---|---|---|
| Reduced carbon to protometabolism (Campaign II) | An autocatalytic motif emerges — OAA self-amplifies ~60× (gain ~3 steady, ~29 peak) via the lumped reductive-TCA turn; a no-CO2 control collapses it. | `gain 29 peak / ~3 steady; control gain 0` |
| Reduced carbon to protometabolism (Campaign II) | Products form ONE connected reaction network (networkx connectivity), not disconnected one-offs. | `connected=1.0` |
| **DILUTION TO LOCALIZED INDIVIDUALS (CAMPAIGN III)** | | |
| Dilution to localized individuals (Campaign III) | A favorable serpentinite pore (5 mm × 0.1 mm, ΔT 30 K, convective flow) reproduces Baaske-2007 extreme accumulation — ~10^8-fold enrichment (accumulation Péclet 18.4) — from the real thermophoretic + convective transport, with FVM flux conserved to machine eps | `favorable Pe_acc 18.4 (enrichment ~9.6e7); residual ~2e-14 dimensionless` |
| Dilution to localized individuals (Campaign III) | A localized individual forms only above a geometric threshold — an open pore (porosity→1) washes the feed through (enrichment ~1) and a 10× shorter pore enriches >100× less, mapping the washout region against the accumulation region. | `open-pore enrichment ~1; short pore >100x less than favorable` |
| **MINERAL COMPARTMENTS TO PROTOCELLS (CAMPAIGN IV)** | | |
| Mineral compartments to protocells (Campaign IV) | Mixed single-chain amphiphiles form viable vesicles at alkaline vent pH and seawater ionic strength, where a single fatty acid does not — reproducing the Jordan-2019 broadening of the pH / salt / divalent stability window. | `mixed viable ~8000; single-fatty-acid viable 0 at 0.35 M salt dimensionless` |
| Mineral compartments to protocells (Campaign IV) | The viable compartments retain internal chemistry (retention 0.91) while staying permeable to the external gradient (access 1.0) — a non-empty individuality-without-isolation region (product ≈ 0.90), the key requirement for a protocell that does not cut itself | `retention 0.91 × access 1.0 ≈ 0.90` |
| **ENVIRONMENTAL TO INTERNAL BIOENERGETICS (CAMPAIGN V)** | | |
| Environmental to internal bioenergetics (Campaign V) | As the external geochemical gradient declines over the run, replicator selection drives the population from externally-driven (pump fraction ~0) to internalized gradient generation (pump fraction ~0.95) while persisting — the bioenergetic internalization is an | `internalization 0.00 -> 0.95; survivors 1.0 at the collapsed gradient dimensionless` |
| Environmental to internal bioenergetics (Campaign V) | A weak pump cannot internalize — when the external gradient collapses, the weak-pump lineage goes extinct (survivors → 0) while the efficient-pump lineage persists. | `efficient survivors 1.0 vs weak-pump survivors 0.0 at collapsed gradient` |
| **CHEMISTRY TO HEREDITY (CAMPAIGN VI)** | | |
| Chemistry to heredity (Campaign VI) | A high-fidelity heredity module (RNA-like, 30-mer, per-site fidelity 0.99 → whole-sequence Q=0.74) retains functional information (C/(C+E)=0.62) and achieves net-positive lineage fitness (0.38) after its replication cost, outscoring the non-heritable control w | `RNA-like fitness 0.38 (retention 0.62) vs non-heritable ~0 dimensionless` |
| Chemistry to heredity (Campaign VI) | Heredity modules rank by an evidence-based head-to-head — RNA-like (0.38) beats the cheaper, lower-fidelity peptide module (0.24), and both beat non-inheritance — while a module below the error threshold (per-site 0.90 × 30-mer, Q=0.04) suffers the error catas | `RNA-like 0.38 > peptide 0.24; low-fidelity retention 0, fitness -0.10` |

## Contradictions & uncertainty

> COMPETING FRAMEWORK **— Classical alkaline-vent (Russell/Martin/Lane/Sojo)**
>
> primary-lens

> COMPETING FRAMEWORK **— 2026 unfavorable reassessment (Tutolo; Jackson)**
>
> challenger

COMPETING FRAMEWORK — **Shallow-vent (Barge & Price)**

alternative-environment

COMPETING FRAMEWORK — **Terrestrial geothermal (Mulkidjanian)**

alternative-environment

COMPETING FRAMEWORK — **Cyanosulfidic/photochemical (Sutherland)**

alternative-environment

CAUSAL CONTINUITY

These studies take their inputs from outside the chain (a supplied feed or reset state) rather than from an upstream study's output — so their result does not yet rest on a modeled upstream mechanism:

Vent reactor baseline (Campaign 0)

### Next studies & priorities

P1 **Ecology before LUCA (Campaign VII)**

Run heterogeneous populations over shared spatial fields; compare emergent capabilities to LUCA reconstructions as a constraint.

— og-07-ecology-before-luca

P1 **Escape from the vent (Campaign VIII)**

Relax environmental support via ablation sweeps and identify the minimum innovation set for autonomy.

— og-08-escape-from-the-vent

P1 **Adversarial: generic nonequilibrium control**

Run the og-01..03 behavior tests on a vent-feature-ablated control; the campaign's gates must reject it.

— og-adv-generic-nonequilibrium

P2 **og-00b** — og-00-vent-reactor-baseline

P2 **og-01b** — og-01-geology-to-reduced-carbon

P2 **og-02b** — og-02-reduced-carbon-to-protometabolism

P2 **og-03b** — og-03-dilution-to-localized-individuals

P2 **og-04b** — og-04-mineral-compartments-to-protocells

P2 **og-05b** — og-05-environmental-to-internal-bioenergetics

P2 **og-06b** — og-06-chemistry-to-heredity

OPEN EXPERT DECISIONS (2)

1. Confirm the first-publication milestone stays og-00 → og-03 (pre-genetic: a precipitating interface that retains a persistent protometabolic carbon flux), or extend to og-04 (protocells)?
2. Heredity strategy for og-06: which modules to prioritize (RNA-world, peptide-first, compositional, mineral-templating)?

## Validation & reproducibility

Automated, code-computed consistency checks (no AI): conservation/invariant ledgers, computed-vs-authored verdicts, run reproducibility (environment + determinism), parameter drift and acceptance-criterion coverage. Each study's own ledger predicates (e.g. mass/charge balance) are its gating behavior tests, listed on the study's card (its Tests).

✓ No open validation flags — the deterministic scan found nothing to act on.

▼ REPORT READINESS — 102 FINDINGS ACROSS 17 CHECKS (27 ERRORS, 75 WARNINGS)

10 **visualization class misplaced** error

`og-00-vent-reactor-baseline, og-01-geology-to-reduced-carbon, og-02-reduced-carbon-to-protometabolism, og-03-dilution-to-localized-individuals +3`

8 **missing provenance** error

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity`

**4 incomplete summaries** error

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity`

**3 missing visualizations** error

`og-07-ecology-before-luca, og-08-escape-from-the-vent, og-adv-generic-nonequilibrium`

**2 readout contract** error

`og-adv-generic-nonequilibrium`

**21 contract-observable** warning

`og-00-vent-reactor-baseline, og-01-geology-to-reduced-carbon, og-02-reduced-carbon-to-protometabolism, og-03-dilution-to-localized-individuals +2`

**12 gate class missing** warning

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity +3`

**9 band test missing cites** warning

`og-00-vent-reactor-baseline, og-01-geology-to-reduced-carbon, og-02-reduced-carbon-to-protometabolism, og-03-dilution-to-localized-individuals +1`

**7 dag edges legacy only** warning

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity +3`

**4 missing conditions block** warning

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity`

**4 authored confidence diverges** warning

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity`

**4 missing simulation set** warning

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity`

**4 missing variants** warning

`og-03-dilution-to-localized-individuals, og-04-mineral-compartments-to-protocells, og-05-environmental-to-internal-bioenergetics, og-06-chemistry-to-heredity`

**3 finding missing cites** warning

`og-00-vent-reactor-baseline, og-02-reduced-carbon-to-protometabolism`

**3 unresolved composite** warning

`og-07-ecology-before-luca, og-08-escape-from-the-vent, og-adv-generic-nonequilibrium`

**3 visualization gap** warning

`og-07-ecology-before-luca, og-08-escape-from-the-vent, og-adv-generic-nonequilibrium`

**1 investigation narrative spine required** warning

`origins-of-life-vent`

## Claim–evidence map

Each study's central scientific claim, its evidence status, and a link to the study.

| CLAIM | Status | Study |
|---|---|---|
| A real PHREEQC + FiPy serpentinization reactor settles into the Lost City window (T 59 °C, pH 9.46) with sustained H2 and exact mass/charge/spatial-flux conservation. | SUPPORTED | Vent reactor baseline (Campaign 0) |
| H2-driven CO2 reduction over Fe/Ni/S reproduces the Preiner/Hudson lab yields and net organic-carbon flux in a bounded favorable regime; the Tutolo-2026 arm shows the usable proton-motive gradient collapses at in-situ T/P, zeroing the flux. | SUPPORTED | Geology to reduced carbon (Campaign I) |
| A curated rTCA-fragment network under continuous feed + washout sustains nonzero carbon throughput, exhibits an autocatalytic motif (OAA self-amplification, gain ~3), and forms a single connected reaction network — metabolism-like behavior emerging without imposed product accumulation. | SUPPORTED | Reduced carbon to protometabolism (Campaign II) |
| A real FiPy thermal trap (thermophoresis + convection) concentrates a dilute vent feed >10^8-fold in a favorable pore (Baaske 2007), and the concentration collapses to ~1 in an open pore — the geometric threshold at which a localized "individual" forms. | SUPPORTED | Dilution to localized individuals (Campaign III) |

| CLAIM | Status | Study |
|---|---|---|
| Mixed single-chain amphiphiles form viable vesicles across the alkaline / seawater-salt window where single fatty acids fail (Jordan 2019), and those compartments retain internal chemistry while staying permeable to the gradient — a non-empty individuality-without-isolation region. | INCONCLUSIVE | Mineral compartments to protocells (Campaign IV) |
| Under a declining external gradient, selection drives an efficient-pump population to internalize gradient generation (pump fraction 0→0.95) and persist, while a weak-pump lineage collapses — the Sojo-2014 leaky-membrane constraint on bioenergetic internalization. | SUPPORTED | Environmental to internal bioenergetics (Campaign V) |
| An inherited (replicable) polymer state improves lineage fitness above its replication cost IF fidelity beats the Eigen error threshold — RNA-like (long, high-fidelity) wins a head-to-head against a cheaper peptide module, and both beat the non-heritable control; below the threshold information is l | SUPPORTED | Chemistry to heredity (Campaign VI) |

## Evidence register

Every study in the campaign (10) — scientific verdict, run status, and one-line result, each linked to its full study card above.

| STUDY | Verdict | Run | Result |
|---|---|---|---|
| Vent reactor baseline (Campaign 0) | SUPPORTED | ran | A real PHREEQC + FiPy serpentinization reactor settles into the Lost City window (T 59 °C, pH 9.46) with sustained H2 and exact mass/charge/spatial-flux conservation. |
| Geology to reduced carbon (Campaign I) | SUPPORTED | ran | H2-driven CO2 reduction over Fe/Ni/S reproduces the Preiner/Hudson lab yields and net organic-carbon flux in a bounded favorable regime; the Tutolo-2026 arm shows the usable proton-motive gradient collapses at in-situ T/ |
| Reduced carbon to protometabolism (Campaign II) | SUPPORTED | ran | A curated rTCA-fragment network under continuous feed + washout sustains nonzero carbon throughput, exhibits an autocatalytic motif (OAA self-amplification, gain ~3), and forms a single connected reaction network — metab |
| Dilution to localized individuals (Campaign III) | SUPPORTED | ran | A real FiPy thermal trap (thermophoresis + convection) concentrates a dilute vent feed >10^8-fold in a favorable pore (Baaske 2007), and the concentration collapses to ~1 in an open pore — the geometric threshold at whic |
| Mineral compartments to protocells (Campaign IV) | INCONCLUSIVE | ran | Mixed single-chain amphiphiles form viable vesicles across the alkaline / seawater-salt window where single fatty acids fail (Jordan 2019), and those compartments retain internal chemistry while staying permeable to the |
| Environmental to internal bioenergetics (Campaign V) | SUPPORTED | ran | Under a declining external gradient, selection drives an efficient-pump population to internalize gradient generation (pump fraction 0→0.95) and persist, while a weak-pump lineage collapses — the Sojo-2014 leaky-membrane |
| Chemistry to heredity (Campaign VI) | SUPPORTED | ran | An inherited (replicable) polymer state improves lineage fitness above its replication cost IF fidelity beats the Eigen error threshold — RNA-like (long, high-fidelity) wins a head-to-head against a cheaper peptide modul |
| Ecology before LUCA (Campaign VII) | PENDING | planned | Run heterogeneous populations over shared spatial fields; compare emergent capabilities to LUCA reconstructions as a constraint. |
| Escape from the vent (Campaign VIII) | PENDING | planned | Relax environmental support via ablation sweeps and identify the minimum innovation set for autonomy. |
| Adversarial: generic nonequilibrium control | PENDING | planned | Run the og-01..03 behavior tests on a vent-feature-ablated control; the campaign's gates must reject it. |

## References

The papers behind each acceptance band, finding and literature anchor.

**Baaske 2007** `baaske2007extreme` — Extreme accumulation of nucleotides in simulated hydrothermal pore systems · Baaske, Philipp and others, 2007 · source ↗ 8×

**Jordan 2019** `jordan2019promotion` — Promotion of protocell self-assembly from mixed amphiphiles at the origin of life · Jordan, Sean F. and others, 2019 · source ↗ 8×

**Sojo 2014** `sojo2014bioenergetic` — A bioenergetic basis for membrane divergence in archaea and bacteria · Sojo, Victor and Pomiankowski, Andrew and Lane, Nick, 2014 · source ↗ 8×

**Huber 1998** `huber1998peptides` — Peptides by activation of amino acids with CO on (Ni,Fe)S surfaces: implications for the origin of life · Huber, Claudia and Wächtershäuser, Günter, 1998 · source ↗ 7×

**Lane 2012** `lane2012origin` — The origin of membrane bioenergetics · Lane, Nick and Martin, William F., 2012 · source ↗ 6×

**Muchowska 2019** `muchowska2019synthesis` — Synthesis and breakdown of universal metabolic precursors promoted by iron · Muchowska, Kamila B. and Varma, Sreejith J. and Moran, Joseph, 2019 · source ↗ 6×

**Preiner 2020** `preiner2020hydrogen` — A hydrogen-dependent geochemical analogue of primordial carbon and energy metabolism · Preiner, Martina and others, 2020 · source ↗ 6×

**Hudson 2020** `hudson2020co2` — CO2 reduction driven by a pH gradient · Hudson, Reuben and others, 2020 · source ↗ 4×

**Kelley 2001** `kelley2001offaxis` — An off-axis hydrothermal vent field near the Mid-Atlantic Ridge at 30 degrees N · Kelley, Deborah S. and others, 2001 · source ↗ 4×

**Ludwig 2006** `ludwig2006formation` — Formation and evolution of carbonate chimneys at the Lost City Hydrothermal Field · Ludwig, Kristin A. and others, 2006 · source ↗ 4×

**Lane 2010** `lane2010chemiosmotic` — How did LUCA make a living? Chemiosmosis in the origin of life · Lane, Nick and Allen, John F. and Martin, William, 2010 · source ↗ 3×

**Sutherland 2017** `sutherland2017opinion` — Studies on the origin of life - the end of the beginning · Sutherland, John D., 2017 · source ↗ 3×

**Tutolo 2026** `tutolo2026rethinking` — Rethinking the origin of life at seafloor hydrothermal vents · Tutolo, Benjamin M., 2026 · source ↗ 3×

**Weiss 2016** `weiss2016physiology` — The physiology and habitat of the last universal common ancestor · Weiss, Madeline C. and others, 2016 · source ↗ 3×

**Herschy 2014** `herschy2014originreactor` — An origin-of-life reactor to simulate alkaline hydrothermal vents · Herschy, Barry and others, 2014 · source ↗ 2×

**Mast 2010** `mast2010thermal` — Thermal trap for DNA replication · Mast, Christof B. and Braun, Dieter, 2010 · source ↗ 2×

**Moody 2024** `moody2024nature` — The nature of the last universal common ancestor and its impact on the early Earth system · Moody, Edmund R. R. and others, 2024 · source ↗ 2×

**Muchowska 2017** `muchowska2017metals` — Metals promote sequences of the reverse Krebs cycle · Muchowska, Kamila B. and others, 2017 · source ↗ 2×

**Schwander 2023** `schwander2023serpentinization` — Serpentinization as the source of energy, electrons, organics, catalysts, nutrients and pH gradients for the origin of LUCA and life · Schwander, Timothy and others, 2023 · source ↗ 2×

**Buckel 2013** `buckel2013energy` — Energy conservation via electron bifurcating ferredoxin reduction and proton/Na+ translocating ferredoxin oxidation · Buckel, Wolfgang and Thauer, Rudolf K., 2013 · source ↗ 1×

**Früh-Green 2003** `fruhgreen2003thirtythousand` — 30,000 years of hydrothermal activity at the Lost City vent field · Früh-Green, Gretchen L. and others, 2003 · source ↗ 1×

**Jackson 2016** `jackson2016natural` — Natural pH gradients in hydrothermal alkali vents were unlikely to have played a role in the origin of life · Jackson, J. Baz, 2016 · source ↗ 1×

Leave an overall note, and per-study notes on each study card. Then **Review & export** assembles them into a YAML feedback report.

FEEDBACK · THE WHOLE REPORT 👍 Looks right ✏️ Needs revision Add a note...

investigations/origins-of-life-vent/investigation.yaml · studies/*/study.yaml · loop-trajectory JSON